Types of Types in the Eventual Categoricity Proof

AEC and EM Reading Notes

Reading notes

Review draft, November 2026

Abstract

These reading notes reorganize the OpenAI eventual categoricity manuscript around a sequence of notions of type: presentation records and order types, automorphism orbits, arrow types, Galois types, and natural or placed canonical diagrams. We emphasize the role of EM functoriality and carefully separate equalities that are proved, implications that are only one-way, and analogies with existing AEC technology. The final sections record locality, compactness, amalgamation at closed levels, and the first definitions for finite diagram categories. This is not an independent verification of the manuscript.

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1 Strategy and notation

Fix an AEC $(\mathbf{K}, \leq_{\mathbf{K}})$ categorical in unboundedly many cardinals. Write $\lambda_0 = \mathrm{LS}(\mathbf{K})$; use λ, κ, μ for infinite cardinals and $\ell(\mathbf{a})$ only for the length of a tuple. Fix an EM template Φ in $\tau \supseteq \tau(\mathbf{K})$. Unless otherwise stated, $\mathrm{EM}_\tau(I, \Phi)$ denotes its reduct to $\tau(\mathbf{K})$.

The argument moves from order types and EM records through automorphism orbits, arrow comparisons and Galois types, toward locality and compactness for placed comparisons, amalgamation at selected cardinals and finite-diagram geometry.

Categoricity in unboundedly many cardinals
$\Downarrow$
EM orbit bounds and bounded-tuple reversal
$\Downarrow$
Stabilized arrow types / canonical diagrams
$\Downarrow$
Locality and compactness for placed comparisons
$\Downarrow$
Amalgamation at selected closed levels
$\Downarrow$
Finite-system comparisons and geometric analysis

Running notation and translation table

Notation	Meaning and reference
$\lambda_0 = \mathrm{LS}(\mathbf{K})$	AEC Löwenheim–Skolem cardinal; replaces the source's cardinal ℓ.
$\ell(\mathbf{a})$, θ, σ	Tuple length and length bounds (Section 2).
$\mathrm{Rec}^\gamma_\tau(I, \Phi)$, ev_I	Presentation records and evaluation (Definition 2.1).
$\mathrm{tp}, \mathrm{tp}^*$	Quantifier-free order type, with and without slot occupancy (Definition 2.2).
$\mathrm{tp}_{\mathrm{aut}}, \mathrm{tp}^{\mathrm{EM}}_{\mathrm{aut}}$	Full and order-induced automorphism types (Section 2.2).
$b_{\mathbf{K}}(\theta)$	Eventual reversal threshold (Theorem 2.10; source Theorem 3.11).
$\equiv^{*,\gamma}_\kappa, \equiv^\gamma_\kappa$	Arrow comparisons (Section 2.3).

Notation	Meaning and reference
$S^{\gamma}_{\kappa}, \mathrm{atp}_{\kappa}, \mathrm{atp}$	Arrow types and their stabilized value (source Proposition 4.2).
$c_{\sigma}, t(\sigma)$	Stabilization and diagram thresholds (source Proposition 4.2).
$\mathrm{gtp}, \mathrm{gtp}^{\mathrm{EM}}$	Ordinary and EM-lifted Galois comparisons (Section 2.4).
$\approx_{\geq \kappa}$	Galois equivalence restricted to intermediate models of size at least κ (Proposition 2.23).
$\mathrm{atp}_{\mathrm{nat}}(I, r)$, $\mathrm{atp}_{\mathrm{nat}}(\mathbf{a}; M)$	Natural arrow type computed from a record, and its record-independent tuple notation (Proposition 2.33).
D_{pl}	Placed diagram over an EM-presented base (source Definition 4.4).
$S^{\sigma}_{\mathrm{atp}_{\mathrm{nat}}}(X; A)$	Natural arrow types of σ-tuples over a parameter subset $X \subseteq A = \mathrm{EM}_{\tau}(I, \Phi)$ (Section 3.2).
$g(\sigma)$	Uniform testing threshold (Section 3.2; source Proposition 4.8).

Remark 1.1 (Notation crosswalk)**.** *The earlier strong type under skeleton-induced automorphisms becomes* $\mathrm{tp}^{\mathrm{EM}}_{\mathrm{aut}}$*. The earlier canonical diagram is the stabilized arrow type* atp*. The ordinary unrestricted Galois type* gtp *can be strictly coarser than* atp *in the absence of amalgamation (Example 2.26). The source's stable diagrams means* stabilized *rather than stable in the AEC sense. Our* $\mathrm{atp}_{\mathrm{nat}}$ *is precisely the source's natural diagram* D_{nat}*; we change the name to emphasize that it is computed from stabilized arrow types. We use* $\mathrm{gtp}^{\mathrm{EM}}$ *only after defining it, not as an automatic synonym for* $\mathrm{atp}_{\mathrm{nat}}$*.*

2 Types of types

We organize the arguments of the source's Sections 3–4 by the kinds of type that are compared. The class $\mathbf{K}^{\mathrm{or}}$ of linear orders with order embeddings provides the index-side invariants; the EM functor lifts order automorphisms, embeddings, and amalgams to $\mathbf{K}$. These give different comparisons of the evaluated tuples.

2.1 Presentation records and order types

Definition 2.1 (Presentation record)**.** *Let I be a linear order. For an output ordinal γ, a* presentation record from I in τ *is a pair $r = (\bar{f}, \mathbf{i})$, where*

$$\bar{f} = (f_k)_{k<\gamma}, \qquad \mathbf{i} = (i_j)_{j<\alpha}$$

is strictly increasing in I, and every f_k is a τ-term in the variables $(x_j)_{j<\alpha}$ using only finitely many variables. Discarding unused indices gives a canonical support convention. The collection $\mathrm{Rec}^{\gamma}_{\tau}(I, \Phi)$ of all such records of fixed output length γ is a set; so is $\mathrm{Rec}^{\leq\theta}_{\tau}(I, \Phi)$ for any fixed cardinal θ. The union over all possible output ordinals would not be a set.

The evaluation map *is*

$$\mathrm{ev}^{\gamma}_{I} : \mathrm{Rec}^{\gamma}_{\tau}(I, \Phi) \longrightarrow \mathrm{EM}_{\tau}(I, \Phi)^{\gamma}, \qquad (\bar{f}, \mathbf{i}) \longmapsto (f_k(\mathbf{i}))_{k<\gamma}.$$

A record r is a presentation record of $\mathbf{a}$ *if $\mathrm{ev}^{\gamma}_{I}(r) = \mathbf{a}$. Every tuple has one by the EM finite-support property, but it need not be unique.*

Definition 2.2 (Order types)**.** *Let* $\mathbf{i}$ *be a sequence from a linear order* I*. Its* quantifier-free order type $\mathrm{tp}(\mathbf{i}; I)$ *records all order and equality relations among its coordinates. The augmented order type* $\mathrm{tp}^*(\mathbf{i}; I)$ *also records whether each strict slot over the distinct coordinates of* $\mathbf{i}$ *is occupied (including the two end slots). For example, where* $x < y$ *are consecutive distinct entries, it records the truth of* $\exists z\,(x < z < y)$*; for infinite tuples this also includes cuts not necessarily specified by a single pair of consecutive entries.*

2.2 Automorphism types

Definition 2.3 (Automorphism orbit type)**.** *For any structure* M *and tuple* $\mathbf{a} \in M^\gamma$*,*

$$\mathrm{tp}_{\mathrm{aut}}(\mathbf{a}; M) = [\mathbf{a}]_{\mathrm{Aut}(M)}$$

is the orbit of $\mathbf{a}$ *under the full automorphism group of* M*. This depends on the ambient model. It is especially useful in homogeneous structures and monster models. For a chosen EM presentation, the subgroup induced by order automorphisms gives the potentially finer* $\mathrm{tp}^{\mathrm{EM}}_{\mathrm{aut}}$ *defined next.*

Remark 2.4 (The homogeneous skeletons)**.** *The source constructs* $\Gamma_\delta = \mathbb{Z}^{(\delta)}$ *and* $H(\delta) = \mathbb{Q}^{(\Gamma_\delta)}$*, using finitely supported sequences and the prescribed reverse-lexicographic orders. For infinite limit* δ *these orders have controlled orbits of short tuples; the key invariant is* tp^**, not merely* tp*.*

Theorem 2.5 (Order automorphism classification [2, Theorem 3.5])**.** *For tuples* $\mathbf{i}, \mathbf{j}$ *of length at most* θ*, where* δ *is an infinite limit ordinal,* θ *is infinite, and* $\theta < \mathrm{cf}(\delta)$*,*

$$\mathrm{tp}^*(\mathbf{i}; H(\delta)) = \mathrm{tp}^*(\mathbf{j}; H(\delta)) \quad \Longleftrightarrow \quad \exists h \in \mathrm{Aut}(H(\delta))\ \big(h(\mathbf{i}) = \mathbf{j}\big).$$

In particular, these order types classify the automorphism orbits of tuples in $H(\delta)$*. For bounded tuple lengths the number of such types is bounded independently of* δ*.*

Definition 2.6 (EM-induced automorphism types)**.** *Fix an infinite cardinal* θ *and suppose* $\mathrm{cf}(\delta) > \theta$*. Write* $M_\delta = \mathrm{EM}_\tau(H(\delta), \Phi)$ *for the* $\tau(\mathbf{K})$*-reduct, with its specified expanded EM presentation understood. Every order automorphism* $h \in \mathrm{Aut}(H(\delta))$ *induces a uniquely specified automorphism* $\widehat{h} = \mathrm{EM}_\tau(h, \Phi) \in \mathrm{Aut}(M_\delta)$ *by functoriality, acting on terms by*

$$\widehat{h}(f(\mathbf{i})) = f(h(\mathbf{i})).$$

For θ*-tuples* $\mathbf{a}, \mathbf{b} \in M_\delta$*, define*

$$\mathbf{a} \equiv^{\mathrm{EM}}_{\mathrm{aut}} \mathbf{b} \quad \Longleftrightarrow \quad \exists h \in \mathrm{Aut}(H(\delta))\ \big(\widehat{h}(\mathbf{a}) = \mathbf{b}\big).$$

This is an equivalence relation, as the induced maps form a group action. Define

$$\mathrm{tp}^{\mathrm{EM}}_{\mathrm{aut}}(\mathbf{a}; M_\delta) := [\mathbf{a}]_{\equiv^{\mathrm{EM}}_{\mathrm{aut}}}, \qquad S^{\mathrm{EM,aut}}_\delta := \{\mathrm{tp}^{\mathrm{EM}}_{\mathrm{aut}}(\mathbf{a}; M_\delta) : \mathbf{a} \in M^\theta_\delta\}.$$

We suppress θ *and* Φ *in* $S^{\mathrm{EM,aut}}_\delta$*. The induced lift* $\widehat{h}$ *is canonical for the fixed EM presentation; the definition is* relative to that presentation *and need not be intrinsic to an abstract copy of the AEC model. In particular, this equivalence can be finer than the full automorphism-orbit equivalence of* M_δ*.*

Lemma 2.7 (Equal occupied-slot records imply equal EM-automorphism type)**.** *Suppose* $\mathbf{a}, \mathbf{b} \in M^\theta_\delta$ *admit presentation records in common. Then* $\mathbf{a} \equiv^{\mathrm{EM}}_{\mathrm{aut}} \mathbf{b}$*.*

Proof. Choose presentations

$$\mathbf{a} = (f_k(\mathbf{i}) : k < \theta), \qquad \mathbf{b} = (f_k(\mathbf{j}) : k < \theta)$$

with the same term list and $\mathrm{tp}^*(\mathbf{i}; H(\delta)) = \mathrm{tp}^*(\mathbf{j}; H(\delta))$. By Theorem 2.5, since the flattened index lists have length at most $\theta < \mathrm{cf}(\delta)$, there is $h \in \mathrm{Aut}(H(\delta))$ with $h(\mathbf{i}) = \mathbf{j}$. Functoriality gives

$$\widehat{h}(\mathbf{a}) = (f_k(h(\mathbf{i})) : k < \theta) = (f_k(\mathbf{j}) : k < \theta) = \mathbf{b}.$$

Thus $\mathbf{a} \equiv^{\mathrm{EM}}_{\mathrm{aut}} \mathbf{b}$. □

Remark 2.8 (Records and orbit bounds)**.** *The converse is not asserted: different term presentations, even of the same tuple, may give different records. This lemma is the direction needed to bound the number of EM-induced automorphism types by the number of possible records. The manuscript's Lemma 3.6 establishes a uniform bound on* full model-automorphism orbits*; the present EM-induced types are orbits under the subgroup induced from the index order, and the same record bound applies when* $\mathrm{cf}(\delta) > \theta$*.*

Definition 2.9 (The arrow relation)**.** *For* $p, q \in S^{\mathrm{EM,aut}}_{\delta}$*, write*

$$p \longrightarrow q \iff \begin{array}{c} \exists \mathbf{a} \models p\ \exists \mathbf{b} \models q\ \exists f : M_\delta \longrightarrow M_\delta \\ f \text{ a strong embedding of the AEC reduct and } f(\mathbf{a}) = \mathbf{b}. \end{array}$$

This relation is well-defined on the EM-induced automorphism types: changes of representatives are effected by induced automorphisms, which are strong automorphisms of the reduct. It is reflexive and transitive; symmetry is the eventual reversal phenomenon discussed below.

The strong-map preorder and presentation profiles. Here the arrow notation is also used for the related but *different* orbit space of an AEC model; this does not identify fixed-skeleton strong types with all model orbits. At a categoricity cardinal κ, fix a model M_κ of size κ. Distinct from the induced-order-automorphism types $S^{\mathrm{EM,aut}}_{\delta}$ just defined, for automorphism orbits p, r of tuples of length at most θ, write

$$p \longrightarrow r$$

if a strong self-embedding of M_κ sends a representative of p to a representative of r. This relation is reflexive and transitive, but symmetry is not automatic.

The paper names the boundedly many full automorphism orbits via maps G_κ and assigns orbit names to presentation records via $F_{\kappa,\delta}$. A presentation *profile* records (i) which records realize which orbit names and (ii) the arrow relation between named orbits. There are boundedly many such profiles.

Lemmas 3.7, 3.8, and 3.10 provide order/ordinal bookkeeping; Proposition 3.9 is the important alignment statement allowing small and large presentations to have matching profile data.

Eventual reversal

Theorem 2.10 (Eventual reversal [2, Theorem 3.11])**.** *For every infinite cardinal* θ *there is a cardinal* $b_{\mathbf{K}}(\theta)$ *such that, whenever* $\kappa \geq b_{\mathbf{K}}(\theta)$ *is a categoricity cardinal,* $M \in \mathbf{K}$ *has size* κ*,* $f : M \to M$ *is strong, and* $\bar{a}$ *has length at most* θ*, there is a strong* $g : M \to M$ *satisfying*

$$g(f(a_i)) = a_i \qquad \text{for every coordinate } i.$$

Equivalently, $\longrightarrow$ becomes symmetric on bounded tuple orbits at sufficiently large categoricity cardinals.

Proof sketch. If asymmetry persisted unboundedly, one could choose repeated profiles at separated categoricity sizes, align small submodels with large presentations, and carry out an elimination construction. The resulting categoricity model would omit the ability to reach a chosen orbit, contradicting the presence of that orbit in the unique model. □

Remark 2.11. Interpretation. *Strong self-embeddings become reversible* on prescribed bounded tuples*; the theorem does not say that every strong self-embedding is an automorphism. This is the first major new structural theorem in our reading.*

2.3 Arrow types

Fix an infinite tuple-length bound θ and a categoricity cardinal $\kappa \geq \lambda_0 + \theta$ above the reversal threshold $b_{\mathbf{K}}(\theta)$ in the definitions below. The existence of a strong embedding carrying one tuple to another is initially directed. The eventual reversal theorem makes it an equivalence relation at sufficiently large categoricity sizes. Its classes are *arrow types*. The stabilized arrow type is precisely what the source calls a *canonical comparison diagram*.

Definition 2.12 (Comparison at size κ)**.** *For $M, N \in \mathbf{K}_\kappa$ and tuples $\mathbf{a} \in M^\theta$, $\mathbf{b} \in N^\theta$, write*

$$(\mathbf{a}; M) \equiv_\kappa^{*,\theta} (\mathbf{b}; N) \quad \Longleftrightarrow \quad \exists f : M \longrightarrow N \text{ strong with } f(\mathbf{a}) = \mathbf{b}.$$

Under the stated hypotheses this is an equivalence relation: reflexivity and transitivity are immediate. For symmetry, use categoricity to identify M and N with the same κ-sized model and apply Theorem 2.10 to the resulting strong self-embedding and the given tuple.

Definition 2.13 (Comparison in larger models and arrow types)**.** *For $M, N \in \mathbf{K}_{\geq\kappa}$ and θ-tuples $\mathbf{a} \in M$, $\mathbf{b} \in N$, write*

$$(\mathbf{a}; M) \equiv_\kappa^{\theta} (\mathbf{b}; N)$$

if there exist $M_0 \preccurlyeq_{\mathbf{K}} M$, $N_0 \preccurlyeq_{\mathbf{K}} N$ of cardinality κ, with $\mathbf{a} \in M_0^\theta$ and $\mathbf{b} \in N_0^\theta$, such that

$$(\mathbf{a}; M_0) \equiv_\kappa^{*,\theta} (\mathbf{b}; N_0).$$

For a tuple $\mathbf{a} \in M^\gamma$, define its arrow type at κ *by*

$$\mathrm{atp}_\kappa(\mathbf{a}; M) := [(\mathbf{a}; M)]_{\equiv_\kappa^\gamma}.$$

The parameter κ refers to the size of the strong submodels used for comparison, not necessarily $|M|$.

Lemma 2.14 (Independence of small submodel [2, Lemma 4.1])**.** *If $M_0, M_1 \preccurlyeq_{\mathbf{K}} M$ have cardinality κ and both contain $\mathbf{a}$, then*

$$(\mathbf{a}; M_0) \equiv_\kappa^{*,\theta} (\mathbf{a}; M_1).$$

Indeed, by downward Löwenheim–Skolem choose a κ-sized $M_* \preccurlyeq_{\mathbf{K}} M$ containing $M_0 \cup M_1$. Coherence yields $M_i \preccurlyeq_{\mathbf{K}} M_*$ for $i = 0, 1$. The inclusions witness $(\mathbf{a}; M_i) \equiv_\kappa^{*,\theta} (\mathbf{a}; M_*)$; symmetry and transitivity give the claim.

Corollary 2.15. *The relation $\equiv_\kappa^\theta$ is an equivalence relation on pairs $(\mathbf{a}; M)$ with $M \in \mathbf{K}_{\geq\kappa}$, and it is independent of which κ-sized strong submodels containing the tuples are chosen. For transitivity, replace the two intermediate κ-submodels inside the middle model by equivalent ones using the lemma, then compose the resulting strong maps at size κ. The relation is preserved by strong embeddings of the ambient models.*

Definition 2.16 (Type sets)**.** *For $\gamma \leq \theta$, set*

$$S_\kappa^\gamma := \{[(\mathbf{a}; M)]_{\equiv_\kappa^\gamma} : M \in \mathbf{K}_{\geq\kappa},\ \mathbf{a} \in M^\gamma\}, \qquad S_\kappa^{\leq\theta} := \coprod_{\gamma\leq\theta} S_\kappa^\gamma.$$

By the small-submodel lemma every class has a representative in a κ-sized model. Categoricity identifies these with the quotient of M_κ^γ by $\equiv_\kappa^{*,\gamma}$, for any fixed $M_\kappa \in \mathbf{K}_\kappa$, so these are sets (up to the usual coding of representatives). The uniform orbit bound controls their cardinalities at sufficiently large categoricity cardinals. The superscript γ denotes the exact *tuple length.*

Remark 2.17 (Arrow types as relative Galois equivalence)**.** *The relation $\equiv_\kappa^\gamma$ compares tuples inside κ-sized strong submodels, even when ambient models are larger. It is the Galois-type equivalence relation* restricted to chains through models of size at least *κ, as made precise in Proposition 2.23 below. It need not coincide with the usual unrestricted Galois equivalence in the absence of amalgamation. Stabilization of these restricted classes is distinct from AEC stability (a bound on types over a model).*

Stabilization of arrow types

Proposition 2.18 (Stabilization [2, Proposition 4.2])**.** *For categoricity cardinals $\lambda \geq \kappa$ above the reversal thresholds, restriction to κ-sized strong submodels gives projections, for each fixed tuple length $\gamma \leq \sigma$,*

$$\pi_{\lambda\kappa}^\gamma : S_\lambda^\gamma \longrightarrow S_\kappa^\gamma, \qquad [(M, \bar{a})] \longmapsto [(M_0, \bar{a})],$$

where $M_0 \leq_{\mathbf{K}} M$ has cardinality κ and contains the tuple. Lemma 2.14 makes this independent of the choice of M_0. Taking the disjoint union gives $\pi_{\lambda\kappa}^{\leq\sigma} : S_\lambda^{\leq\sigma} \to S_\kappa^{\leq\sigma}$. These maps compose and are surjective: an EM presentation $\mathrm{EM}_\tau(I, \Phi)$ at size κ can be extended, by extending I, to a presentation $\mathrm{EM}_\tau(J, \Phi)$ at size λ (with the AEC reduct understood).

More precisely, there is a categoricity cardinal c_σ above the σ-tuple reversal threshold such that for every pair of categoricity cardinals $c_\sigma \leq \kappa \leq \lambda$, the projection $\pi_{\lambda\kappa}^{\leq\sigma}$ is bijective. This is stabilization of the comparison classes*, not model-theoretic stability.*

Proof sketch. Suppose that noninjective projections occur arbitrarily late. Choose B_σ^+ successive refinement stages and a common categoricity cardinal above them. Each refinement produces a pair of classes in the final model agreeing at an earlier stage but distinct at a later one. Pairs arising at different stages are distinct, because separation persists under further projection. Yet the final class set has size at most the infinite cardinal B_σ, and hence has at most $B_\sigma^2 = B_\sigma$ ordered pairs, a contradiction.

□

Definition 2.19 (Stabilized arrow type (canonical diagram); extracted from [2, Proposition 4.2])**.** *Fix an infinite cardinal σ and a categoricity cardinal c_σ beyond the stabilization and reversal thresholds for tuples of length at most σ. For $M \in \mathbf{K}_{\geq c_\sigma}$ and $\bar{a} \in M^\gamma$, $\gamma \leq \sigma$, define*

$$\mathrm{atp}(\bar{a}; M) := [(M_0, \bar{a})]_{\equiv_{c_\sigma}^\gamma},$$

where $M_0 \leq_{\mathbf{K}} M$ has size c_σ and contains $\bar{a}$. Lemma 2.14 makes the definition independent of M_0. Stabilization makes equality of these diagrams independent of the chosen sufficiently large categoricity cardinal. We also write $D_\sigma(\bar{a}; M)$ or $D(\bar{a}; M)$ when quoting the source. Thus the stabilized arrow type atp *is precisely its canonical comparison diagram.*

Proposition 4.2 of the source manuscript further establishes:

(1) Equality of diagrams is preserved by strong embeddings; there are at most B_σ diagrams of the prescribed bounded lengths.

(2) Diagrams contain atomic information and respect restriction, reindexing, and enlargement of the tuple bound.

(3) At categoricity cardinals at least the paper's nondecreasing threshold $t(\sigma)$, equality of diagrams is implemented by a strong map.

(4) Compatible joint diagrams can be transported over any realization of the relevant initial diagram in a sufficiently large categoricity model.

The paper chooses $t(\sigma)$ above the stabilization thresholds c_τ for every infinite $\tau \leq \sigma$; no explicit numerical formula for c_σ is supplied.

The canonical diagram is *not* simply a presentation record from the order classification (Theorem 2.5). Records depend on chosen EM coordinates and give sufficient conditions for orbit equivalence; stabilized strong-map classes are intrinsic to the AEC.

2.4 Galois types and EM-lifted comparisons

The standard AEC Galois type is obtained from joint embedding of tuples, not directly from the choice of a particular EM presentation. Without amalgamation, we use the equivalence relation generated by direct joint-embedding comparisons.

Definition 2.20 (Galois types over the empty set)**.** *For pairs $(\mathbf{a}; M)$ and $(\mathbf{b}; N)$ of equal tuple length, say they are* directly equivalent by amalgamation *if there are strong embeddings $f_0 : M \to P$ and $f_1 : N \to P$ with $f_0(\mathbf{a}) = f_1(\mathbf{b})$. Without assuming amalgamation, define equality of* $\mathrm{gtp}(\mathbf{a}/\emptyset; M)$ *and* $\mathrm{gtp}(\mathbf{b}/\emptyset; N)$ *as the equivalence relation generated by direct equivalence (its transitive closure). This is distinct from the atomic diagram of a tuple and makes sense without assuming the amalgamation property.*

Definition 2.21 (Galois types over a model)**.** *More generally, for $A \leq_{\mathbf{K}} M$ and $\mathbf{a} \in M^\gamma$, the* Galois type of $\mathbf{a}$ over A in M, $\mathrm{gtp}(\mathbf{a}/A; M)$, *is the equivalence class generated by direct identifications of tuples in strong amalgams* over the same copy of A. *Thus $(\mathbf{a}; M)$ and $(\mathbf{b}; N)$ are directly equivalent over A if $A \leq_{\mathbf{K}} M, N$ and there are $\mathbf{K}$-embeddings $f : M \to P$ and $g : N \to P$ with $f \restriction A = g \restriction A$ and $f(\mathbf{a}) = g(\mathbf{b})$. Without amalgamation, we take the transitive closure to obtain equivalence. We abbreviate* $\mathrm{gtp}(\mathbf{a}; M) := \mathrm{gtp}(\mathbf{a}/\emptyset; M)$ *when the parameter base is empty; this is the meaning of* $\mathrm{gtp}(\mathbf{a}; M)$ *in the comparisons here.*

Lemma 2.22 (Direct amalgamation versus strong maps at a categoricity size)**.** *Suppose $\kappa \geq b_{\mathbf{K}}(\theta)$ is a categoricity cardinal with $\kappa \geq \lambda_0 + \theta$. For $M, N \in \mathbf{K}_\kappa$ and tuples $\mathbf{a} \in M^\gamma$, $\mathbf{b} \in N^\gamma$ of length $\gamma \leq \theta$, the following are equivalent:*

(i) *There is a* $\mathbf{K}$*-embedding* $f : M \to N$ *with* $f(\mathbf{a}) = \mathbf{b}$.

(ii) *There are* $\mathbf{K}$*-embeddings* $f_0 : M \to P$ *and* $f_1 : N \to P$ *with* $f_0(\mathbf{a}) = f_1(\mathbf{b})$.

Here (ii) is direct amalgamation equivalence*, not the unrestricted transitive-closure definition of* $\mathrm{gtp}(-/\emptyset; -)$.

Proof. The first condition implies the second by using $P = N$, $f_0 = f$, and $f_1 = \mathrm{id}_N$. For the converse, downward Löwenheim–Skolem and coherence give $P_0 \leq_{\mathbf{K}} P$ of size κ containing $f_0[M] \cup f_1[N]$, through which both embeddings factor strongly. By categoricity, choose $h : P_0 \cong N$. Then $hf_1 : N \to N$ is a $\mathbf{K}$-embedding. Apply eventual reversal to hf_1 and $\mathbf{b}$, obtaining $g : N \to N$ strong with $ghf_1(\mathbf{b}) = \mathbf{b}$. Therefore $ghf_0 : M \to N$ carries $\mathbf{a}$ to $\mathbf{b}$. □

Proposition 2.23 (Arrow types and tail-restricted Galois equivalence)**.** *Under the hypotheses of Lemma 2.22, let* $\approx_{\geq\kappa}$ *be the equivalence relation on pairs* $(\mathbf{a}; M)$ *with* $M \in \mathbf{K}_{\geq\kappa}$ *generated by direct amalgamation identifications* using only ambient and intermediate models of size at least κ*. For* $M, N \in \mathbf{K}_{\geq\kappa}$ *and tuples of the same length* $\gamma \leq \theta$*,*

$$\mathrm{atp}_\kappa(\mathbf{a}; M) = \mathrm{atp}_\kappa(\mathbf{b}; N) \iff (\mathbf{a}; M) \approx_{\geq\kappa} (\mathbf{b}; N).$$

Consequently,

$$\mathrm{atp}_\kappa(\mathbf{a}; M) = \mathrm{atp}_\kappa(\mathbf{b}; N) \implies \mathrm{gtp}(\mathbf{a}/\emptyset; M) = \mathrm{gtp}(\mathbf{b}/\emptyset; N).$$

The converse to the last implication is not asserted without further hypotheses.

Proof. If the arrow types agree, choose κ-sized strong submodels $M_0 \leq_{\mathbf{K}} M$ and $N_0 \leq_{\mathbf{K}} N$ containing the tuples, and a strong embedding $f : M_0 \to N_0$ carrying $\mathbf{a}$ to $\mathbf{b}$. The inclusions and f give a three-step chain of direct identifications through M_0, N_0, all of size at least κ.

Conversely, consider one direct identification between $(\mathbf{a}; M)$ and $(\mathbf{b}; N)$ at sizes at least κ, witnessed in P. Choose κ-sized strong submodels $M_0 \leq_{\mathbf{K}} M$ and $N_0 \leq_{\mathbf{K}} N$ containing the tuples. Their images in P lie in a κ-sized strong submodel $P_0 \leq_{\mathbf{K}} P$ by downward Löwenheim–Skolem and coherence. Lemma 2.22 yields a strong map $M_0 \to N_0$ identifying the tuples, hence equal κ-arrow types. Transitivity of arrow-type equality handles a finite chain of direct identifications. The implication to ordinary Galois types follows because every allowed $\approx_{\geq\kappa}$ comparison is also allowed in the unrestricted relation. □

Corollary 2.24 (Earlier stabilization if amalgamation is assumed)**.** *If* $\mathbf{K}$ *has the amalgamation property, then at every categoricity cardinal* $\kappa \geq b_{\mathbf{K}}(\theta)$ *with* $\kappa \geq \lambda_0 + \theta$*,*

$$\mathrm{atp}_\kappa(\mathbf{a}; M) = \mathrm{atp}_\kappa(\mathbf{b}; N) \iff \mathrm{gtp}(\mathbf{a}/\emptyset; M) = \mathrm{gtp}(\mathbf{b}/\emptyset; N)$$

for models M, N *of size at least* κ *and tuples of length at most* θ*. Hence the comparison projections between any two such categoricity cardinals are already bijective. This conditional conclusion cannot be used in the manuscript before amalgamation has been established.*

Proof. Amalgamation makes direct amalgamation equivalence transitive, so unrestricted Galois-type equality has a direct witness. Restricting that witness to κ-sized strong submodels as in the proof of Proposition 2.23 gives equal arrow types. The converse is the same proposition. For $\lambda \geq \kappa$ as in the statement, both arrow comparisons compute the same unrestricted Galois type, proving injectivity of the projection. Surjectivity follows from the order-presented extensions supplied by the EM functor. □

Remark 2.25 (Why the reversal threshold is not a stabilization threshold)**.** *Ordinary Galois equivalence over $\emptyset$, as defined above without amalgamation, allows the intermediate models in its transitive closure to have* any *cardinality. It can therefore be strictly coarser than $\approx_{\geq\kappa}$, even for κ-sized endpoints. The final sentence of the previous version of Lemma 2.22 incorrectly inferred equality for the unrestricted transitive closure.*

Neither no maximal models *nor the Löwenheim–Skolem property repairs this gap. Consider two direct identifications*

$$M \longrightarrow P \longleftarrow A, \qquad A \longrightarrow Q \longleftarrow N$$

through a small intermediate model A. No maximal models lets us enlarge all three witness models, and Löwenheim–Skolem lets us extract submodels of a prescribed sufficiently large size. But these principles do not produce one extension of A *which embeds compatibly into both P and Q. That is an amalgamation requirement. The star example below has no maximal models and joint embedding, yet exhibits precisely this obstruction.*

As the cutoff κ increases, the relations $\approx_{\geq\kappa}$ can refine one another; Proposition 4.2 of the source proves that the associated arrow classes eventually stabilize by a uniform counting argument. Invariance of unrestricted Galois types under strong embeddings does not by itself lower the stabilization threshold to $b_{\mathbf{K}}(\theta)$. If $\mathbf{K}$ has amalgamation, direct amalgamation equivalence is transitive and the distinction disappears.

Example 2.26 (Stars: a counterexample to unrestricted equivalence)**.** *Let $\mathbf{K}$ have a binary relation R. Its models are singletons and stars with a distinguished center c satisfying $R(c, x)$ for every leaf x, and no other instances of R. Declare every singleton substructure strong, while a nonsingleton substructure is strong precisely when it contains the center of its ambient star. This is an AEC with $\mathrm{LS}(\mathbf{K}) = \aleph_0$, categorical in every infinite cardinal (a model of size κ is a star with κ leaves). It admits the usual EM presentation with one constant for the center and order-indexed leaves.*

In a κ-sized star M, take its center c and a leaf d. Strong self-embeddings preserve the center, so

$$\mathrm{atp}_{\kappa}(c; M) \neq \mathrm{atp}_{\kappa}(d; M).$$

Yet the unique element x of a singleton strong model X can be embedded into M as either c or d. Hence $(c; M)$ is directly equivalent to $(x; X)$ and $(x; X)$ directly equivalent to $(d; M)$, giving

$$\mathrm{gtp}(c/\emptyset; M) = \mathrm{gtp}(d/\emptyset; M).$$

This AEC has no maximal models*: every singleton extends strongly to arbitrarily large stars, and every star can be extended by adding more leaves. It also has joint embedding, since two stars can be embedded into a larger star with their centers identified. Neither property prevents the singleton X from serving as an intermediate model in a Galois-equivalence chain that joins distinct arrow types.*

The reversal conclusion still holds for tuples shorter than κ: a partial inverse of a strong self-embedding on fewer than κ leaves extends to an injection on all the leaves.

Example 2.27 (Colored arrows: arbitrarily long Galois zigzags)**.** *Consider the language with unary T and binary relations $(R_i)_{i<\omega}$. Think of $T(x)$ as saying that x is terminal, and of $R_i(x, y)$ as an outgoing edge of color i. Let $\mathbf{K}_{\mathrm{col}}$ consist of structures satisfying:*

(i) *a vertex satisfying T has no outgoing R_i-edge;*

(ii) *every vertex not satisfying T has at least one outgoing edge;*

(iii) *whenever* $R_i(x,y)$ *and* $R_j(x,z)$ *hold,* $|i-j| \leq 1$.

Take the strong-submodel relation to be induced substructure inclusion between models of $\mathbf{K}_{\mathrm{col}}$*. Directed unions preserve these conditions, and the Löwenheim–Skolem number is* $\aleph_0$*: from any set of elements, iteratively add witnesses for outgoing edges of nonterminal vertices, in countably many steps.*

Fix a terminal singleton $M = \{m\}$*. For each* $i < \omega$*, take a three-element model* $N_i = \{m, a_i, b_i\}$ *with* m, b_i *terminal and* $R_i(a_i, b_i)$ *the unique edge. Write* $p_i = \mathrm{gtp}(a_i/M; N_i)$*, using unrestricted Galois equivalence as in our definitions. More precisely, denote by* $q_i = (a_i, M, N_i)$ *the* pointed triple *before taking its equivalence class.*

The triples q_i, q_{i+1} *are directly equivalent: identify* a_i *with* a_{i+1} *in a common model over* M*, retaining their outgoing edges of adjacent colors. But* q_i *and* q_j *are not directly equivalent when* $|i-j| \geq 2$*: the identified vertex would have outgoing edges of colors differing by at least two. Thus all the* q_i *have the same* Galois type*, since direct equivalence is connected by a finite chain, even though a direct amalgam of the original endpoints need not exist.*

In fact, the shortest chain of direct equivalences from q_0 *to* q_n *has length* n*. For a pointed triple with nonterminal distinguished element, let* $m(q)$ *be the minimum color of an outgoing edge at that element. A direct equivalence implies* $|m(q) - m(q')| \leq 1$*, since both colors must occur at the identified vertex of an amalgam. Therefore a chain from* $m(q_0) = 0$ *to* $m(q_n) = n$ *requires at least* n *steps; the adjacent-color chain attains this bound.*

Comparison with stars. *The star example shows that allowing* small *intermediate models can identify two different arrow types, even with no maximal models. This colored-arrow example illustrates a separate phenomenon: the transitive closure in the definition of Galois type can require arbitrarily long finite chains of direct amalgamations.*

Remark 2.28 (Scope of the characterization)**.** *For ambient* M, N *of size at least* κ*, equality of* κ*-arrow types is equivalent to equality in the* tail-restricted *equivalence* $\approx_{\geq\kappa}$*, not necessarily to equality of unrestricted Galois types. The latter always follows from arrow-type equality but may hold without it, as Example 2.26 shows. This is an issue even when* M, N *themselves have cardinality* κ*: a global Galois-type chain may pass through smaller models.*

We write S^{γ}_{κ} *for the arrow-type classes and* S^{γ}_{*} *for their eventual common value under the projections of Proposition 4.2, understood up to canonical bijections. These classes need not be all global Galois types.*

Definition 2.29 (EM-lifted Galois comparison)**.** *Fix the EM template* Φ*. A* direct EM-lifted comparison *between tuples* $(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi))$ *and* $(\mathbf{b}; \mathrm{EM}_\tau(J, \Phi))$ *consists of presentation records* $r = (\bar{f}, \mathbf{i})$ *of* $\mathbf{a}$ *and* $s = (\bar{f}, \mathbf{j})$ *of* $\mathbf{b}$ *with the same term sequence and*

$$\mathrm{tp}(\mathbf{i}; I) = \mathrm{tp}(\mathbf{j}; J).$$

Equivalently, the two index tuples have the same Galois type over the empty set in $\mathbf{K}^{\mathrm{or}}$*: there is a joint order extension identifying them. The lifted order embeddings yield strong embeddings of EM models identifying the evaluated tuples.*

Write

$$\mathrm{gtp}^{\mathrm{EM}}(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi)) = \mathrm{gtp}^{\mathrm{EM}}(\mathbf{b}; \mathrm{EM}_\tau(J, \Phi))$$

if the two tuples are equivalent under the equivalence relation generated by direct EM-lifted comparisons*. This is an invariant of* EM-presented tuples*, rather than the usual AEC Galois type on*

all $\mathbf{K}$*-models. The use of transitive closure ensures that the notation really denotes an equivalence class even when a tuple has several presentation records.*

Proposition 2.30 (Lifting quantifier-free order types)**.** *Suppose* $\mathbf{a}$ *and* $\mathbf{b}$ *admit records* $(\bar{f}, \mathbf{i})$ *and* $(\bar{f}, \mathbf{j})$ *with* $\mathrm{tp}(\mathbf{i}; I) = \mathrm{tp}(\mathbf{j}; J)$*. Then*

$$\mathrm{gtp}^{\mathrm{EM}}(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi)) = \mathrm{gtp}^{\mathrm{EM}}(\mathbf{b}; \mathrm{EM}_\tau(J, \Phi)),$$

and their natural diagrams coincide.

Proof. Amalgamate I and J over the labeled support isomorphism $\mathbf{i} \mapsto \mathbf{j}$, obtaining order embeddings into a common linear order L. By EM functoriality, the induced $\mathbf{K}$-embeddings of the reducts into $\mathrm{EM}_\tau(L, \Phi)$ identify the tuples. This is a direct EM-lifted comparison. Extending L to a sufficiently large categoricity-order presentation and using preservation of stabilized arrow types proves equality of natural diagrams. □

Remark 2.31 (The comparison under investigation)**.** *The notation* $\mathrm{gtp}^{\mathrm{EM}}$ *in these notes is a relation on* tuples in EM models*, generated by comparisons induced from order embeddings. The natural diagram is also ultimately an invariant of the evaluated tuple, although a record is used to compute it. The genuine difference is the* class of morphisms permitted*: order-induced EM embeddings for* $\mathrm{gtp}^{\mathrm{EM}}$*, but arbitrary strong maps in the stabilized arrow type. The former implies equality of natural diagrams. Example 2.37 below shows that the converse can fail. We retain* $\mathrm{gtp}^{\mathrm{EM}}$ *primarily to exhibit the distinction between the order-induced and the full AEC comparisons; it is a useful auxiliary invariant, but is not expected to play a central role in the subsequent arguments. The working calculus is based on* atp *and its EM-based extension* $\mathrm{atp}_{\mathrm{nat}}$*.*

Natural and placed canonical diagrams

Definition 2.32 (Natural arrow type (source: natural diagram); [2, Definition 4.4])**.** *Fix a linear order* I*, an output ordinal* γ*, and a presentation record* $r = (\bar{f}, \mathbf{i}) \in \mathrm{Rec}^\gamma_\tau(I, \Phi)$*. Choose an order extension* $j : I \hookrightarrow I'$ *whose cardinality is a sufficiently large categoricity cardinal for the stabilized arrow type of a* γ*-tuple. Write* $j_* r = (\bar{f}, j(\mathbf{i}))$ *for the induced record. Define*

$$\mathrm{atp}_{\mathrm{nat}}(I, r) := \mathrm{atp}\big(\mathrm{ev}^\gamma_{I'}(j_* r); \mathrm{EM}_\tau(I', \Phi)\big).$$

This is the natural diagram computed from (I, r)*. The purpose of* I' *is to evaluate the record at a sufficiently large categoricity cardinal where* atp *is defined and stabilized, not to introduce a new type invariant. Neither its choice nor the chosen record affects the result, as proved below.*

Proposition 2.33 (Natural diagrams descend from records to tuples)**.** *Let* I *be a linear order and* $r, s \in \mathrm{Rec}^\gamma_\tau(I, \Phi)$ *be records with*

$$\mathrm{ev}^\gamma_I(r) = \mathrm{ev}^\gamma_I(s) = \mathbf{a}.$$

Then $\mathrm{atp}_{\mathrm{nat}}(I, r) = \mathrm{atp}_{\mathrm{nat}}(I, s)$*. Furthermore, the result of Definition 2.32 is independent of the sufficiently large extension* I'*. Thus it is well-defined to write*

$$\mathrm{atp}_{\mathrm{nat}}(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi)) := \mathrm{atp}_{\mathrm{nat}}(I, r)$$

for any *presentation record* r *of* $\mathbf{a}$*. If* $|\mathrm{EM}_\tau(I, \Phi)|$ *is already above the relevant stabilization threshold, this equals* $\mathrm{atp}(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi))$*.*

Proof. For a common order extension $j : I \hookrightarrow I'$, functoriality of the expanded EM presentation gives

$$\mathrm{ev}^{\gamma}_{I'}(j_* r) = \mathrm{EM}_{\tau}(j, \Phi)(\mathbf{a}) = \mathrm{ev}^{\gamma}_{I'}(j_* s).$$

They therefore have the same stabilized arrow type. For two different sufficiently large extensions I'_0, I'_1, amalgamate the linear orders over the common copy of I, and extend the resulting order if necessary to another sufficiently large categoricity cardinal. The induced strong EM maps identify the evaluated tuple, so preservation of stabilized arrow types makes the two values equal. Finally, if the original reduct is already large enough to carry atp, its strong embedding into any such extension preserves $\mathrm{atp}(\mathbf{a}; -)$. □

Remark 2.34 (Ambient invariance and the role of the EM presentation)**.** *For any fixed large-enough categoricity cardinal κ, $\mathrm{atp}_{\kappa}(\mathbf{a}; M)$ is already preserved by every strong embedding $M \to N$ of models of size at least κ; it does* not *depend on which strong extension of such an M is used. The potential difficulties are instead:*

(i) *the arrow comparisons may refine as κ increases, until their eventual stabilization; and*

(ii) *a small model M may admit* incompatible *strong extensions whose images of $\mathbf{a}$ have different stabilized arrow types. No-maximal-models alone does not provide a well-defined choice of type.*

The special advantage of an EM presentation $\mathrm{EM}_{\tau}(I, \Phi)$ is that order extensions $I \hookrightarrow I'$ *provide a compatible directed family of strong extensions. Since linear orders amalgamate over a common suborder, the stabilized arrow type calculated in this family is independent of the chosen sufficiently large I'. There is no distinguished canonical order I', and the extension itself is neither canonical nor natural in a categorical sense. Instead, the* value of the type *is independent of every permitted choice of I', relative to the fixed template and EM presentation. In this precise sense $\mathrm{atp}_{\mathrm{nat}}$ gives a canonical extension of the* stabilized arrow-type invariant *to tuples in smaller EM-presented models, rather than a canonical extension of the models themselves. For models already above the stabilization threshold, it coincides with the intrinsic* atp*.*

Proposition 2.35 (Order-pattern sufficiency for natural diagrams)**.** *Suppose records $r = (\bar{f}, \mathbf{i})$ over I and $s = (\bar{f}, \mathbf{j})$ over J have the same term list and satisfy $\mathrm{tp}(\mathbf{i}; I) = \mathrm{tp}(\mathbf{j}; J)$. Then*

$$\mathrm{atp}_{\mathrm{nat}}(I, r) = \mathrm{atp}_{\mathrm{nat}}(J, s).$$

No occupancy information about the ambient orders is required. This is a sufficient *criterion, not a characterization of diagram equality.*

Proof. The supports are isomorphic as ordered lists. Amalgamate I and J over that support isomorphism and lift the order embeddings through the EM functor. The resulting strong embeddings identify the evaluated tuples; extend to a sufficiently large categoricity order and use preservation of stabilized arrow types. Equivalently, apply Proposition 2.30. □

Corollary 2.36 (EM-lifted comparison implies natural-diagram equality)**.** *For tuples $\mathbf{a}, \mathbf{b}$ in EM-presented models,*

$$\mathrm{gtp}^{\mathrm{EM}}(\mathbf{a}) = \mathrm{gtp}^{\mathrm{EM}}(\mathbf{b}) \quad \Longrightarrow \quad \mathrm{atp}_{\mathrm{nat}}(\mathbf{a}) = \mathrm{atp}_{\mathrm{nat}}(\mathbf{b}),$$

where the right side abbreviates the record-independent tuple notation of Proposition 2.33.

Proof. Each generating direct EM-lifted comparison is induced by common records and the same quantifier-free order type, so Proposition 2.35 gives equality of natural diagrams. This equality persists along a finite chain of generating comparisons. □

Example 2.37 (The converse fails even for pure sets)**.** *Let* $\mathbf{K}$ *be the AEC of pure sets with strong embeddings the injections, categorical in every infinite cardinal. Take an EM template whose expanded structure* $\mathrm{EM}_\tau(I, \Phi)$ *has underlying set* I*, remembers its linear order* $<$*, and has no functions other than variables as terms. Its reduct is the pure set* I*.*

Choose $i < j$ *in* I *and let* $\mathbf{a} = (i, j)$ *and* $\mathbf{b} = (j, i)$*. In the pure-set reduct, an automorphism interchanges* i *and* j*, hence*

$$\mathrm{atp}_{\mathrm{nat}}(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi)) = \mathrm{atp}_{\mathrm{nat}}(\mathbf{b}; \mathrm{EM}_\tau(I, \Phi)).$$

But every order-induced embedding preserves whether the first tuple coordinate precedes the second in the skeleton. This property is preserved under finite chains of EM-lifted comparisons, so

$$\mathrm{gtp}^{\mathrm{EM}}(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi)) \neq \mathrm{gtp}^{\mathrm{EM}}(\mathbf{b}; \mathrm{EM}_\tau(I, \Phi)).$$

One can present $\mathbf{a}$ *using the term sequence* (x_0, x_1) *and* $\mathbf{b}$ *using* (x_1, x_0) *over the same increasing support* (i, j)*; there are no alternative term presentations that remove the reversal in this template. Thus even the* existence *of records with equal natural diagrams does not imply equality of the EM-lifted type.*

Remark 2.38 (Resolution of the earlier review question)**.** *The tuple-invariant interpretation of* $\mathrm{atp}_{\mathrm{nat}}$ *is legitimate by Proposition 2.33. The distinction from* $\mathrm{gtp}^{\mathrm{EM}}$ *is not dependence on a chosen record: both assign equivalence classes to tuples in their respective EM presentations. However,* $\mathrm{gtp}^{\mathrm{EM}}$ *remembers the chosen order presentation in its permitted witnesses, whereas* $\mathrm{atp}_{\mathrm{nat}}$ *is intrinsic to the evaluated AEC tuple once sufficiently large. The EM-lifted relation uses a more restrictive family of witnesses. Example 2.37 shows that equality of natural diagrams is strictly weaker in general. Accordingly we retain two distinct names rather than define* $\mathrm{gtp}^{\mathrm{EM}} := \mathrm{atp}_{\mathrm{nat}}$*.*

Definition 2.39 (Placement and placed diagram over a base; [2, Definition 4.4])**.** *Fix* $A = \mathrm{EM}_\tau(I, \Phi)$*, an enumeration* $\mathbf{a}_A$ *of its elements, and term representations of that enumeration using positions in* I*. A* placement *of an additional tuple* $\mathbf{d}$ *over* A *is a labeled term representation of* $\mathbf{d}$ *in an order extension* $I \subseteq J$*, retaining the fixed positions and term representations of all parameters in* $\mathbf{a}_A$*. The* placed comparison diagram *of this placement is*

$$D_{\mathrm{pl}}(\mathbf{d};\ A, \textit{placement}) := \mathrm{atp}_{\mathrm{nat}}((\mathbf{d}, \mathbf{a}_A); \mathrm{EM}_\tau(J, \Phi)),$$

where the right side uses the combined labeled term representation. It is computed in a sufficiently large EM extension, which need not have the cardinality of A*.*

Remark 2.40 (Relative order records and cuts)**.** *A placement's order-theoretic data may be described by the quantifier-free order type of the new support coordinates over the fixed skeleton* I*, together with the term labels and the information specifying which positions occur in each term. Equivalently, record the cuts in* I *of the new coordinates and their mutual order/equality relations (especially for coordinates in the same cut). These are cuts of the* skeleton order*, not necessarily cuts of the underlying structure* A*.*

Thus a placement resembles a presentation record with parameters, but its placed diagram *is a* natural diagram with parameters*, not an EM-automorphism orbit presentation record. Matching relative labeled order records implies matching placed diagrams; the converse is not asserted. Nor do we identify placed diagrams with Galois types over arbitrary bases before the relevant amalgamation results have been proved.*

Remark 2.41 (Relation to types over models)**.** *For $A = \mathrm{EM}_\tau(I, \Phi)$ and a tuple $\mathbf{d}$ in an EM extension, the placed comparison is* $\mathrm{atp}_{\mathrm{nat}}((\mathbf{d}, \mathbf{a}_A); \mathrm{EM}_\tau(J, \Phi))$*, with the enumeration of A fixed on both sides. This resembles the usual AEC notion* $\mathrm{gtp}(\mathbf{d}/A; N)$*, but the definition requires neither a monster model nor amalgamation over A. Matching the labeled relative support order pattern is sufficient for equality of placed diagrams, not necessary. Conversely, equality of placed diagrams need not imply that the two chosen presentations have isomorphic labeled support orders.*

Remark 2.42 (Comparisons between the type notions)**.** *For tuples represented by a common term list $\bar{f}$, the implications proved here can be summarized as*

$$\mathrm{tp}^*(\mathbf{i}; H(\delta)) = \mathrm{tp}^*(\mathbf{j}; H(\delta)) \Longrightarrow \mathrm{tp}^{\mathrm{EM}}_{\mathrm{aut}}(\mathbf{a}; M_\delta) = \mathrm{tp}^{\mathrm{EM}}_{\mathrm{aut}}(\mathbf{b}; M_\delta) \Longrightarrow \mathrm{tp}_{\mathrm{aut}}(\mathbf{a}; M_\delta) = \mathrm{tp}_{\mathrm{aut}}(\mathbf{b}; M_\delta).$$

The first implication assumes the usual bound $\ell(\mathbf{i}) < \mathrm{cf}(\delta)$. For possibly distinct index orders we have

$$\begin{aligned} &\mathrm{tp}(\mathbf{i}; I) = \mathrm{tp}(\mathbf{j}; J) \\ &\quad \Longrightarrow \mathrm{gtp}^{\mathrm{EM}}(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi)) = \mathrm{gtp}^{\mathrm{EM}}(\mathbf{b}; \mathrm{EM}_\tau(J, \Phi)) \\ &\quad \Longrightarrow \mathrm{atp}_{\mathrm{nat}}(\mathbf{a}; \mathrm{EM}_\tau(I, \Phi)) = \mathrm{atp}_{\mathrm{nat}}(\mathbf{b}; \mathrm{EM}_\tau(J, \Phi)). \end{aligned}$$

These implications do not assert converses.

At a sufficiently large categoricity cardinal, atp_κ *is exactly tail-restricted Galois equivalence by Proposition 2.23. It implies ordinary Galois-type equality over $\emptyset$, but the converse can fail without amalgamation, even at cardinality κ. Similarly, a placed diagram over $A = \mathrm{EM}_\tau(I, \Phi)$ is an arrow comparison with an enumeration of A included as parameters; it is not declared to equal a Galois type over arbitrary A.*

3 Consequences: order symmetry, locality, and amalgamation

3.1 Back-and-forth and block transpositions

Remark 3.1 (Source Proposition 4.3: back-and-forth)**.** *At a closure point ζ of countable cofinality, write both ζ-models as countable increasing unions of bounded sets. Alternately extend tuples with matching diagrams, using source Proposition 4.2 to obtain strong maps between bounded strong submodels. The union of the resulting partial correspondences is an isomorphism because atomic information is preserved. The argument does not require compatible EM resolutions of the two models. At uncountable cofinality there are intermediate limit stages, where preservation of a possibly infinite matched tuple's diagram needs additional justification.*

Lemma 3.2 (Disjoint block transpositions; [2, Lemma 4.6])**.** *Fix a linear order I and a labeled EM presentation $M = \mathrm{EM}_\tau(I, \Phi)$. Let $\mathbf{d}, \mathbf{e}$ be (possibly set-sized) tuples of elements of M, and choose term presentations*

$$d_\alpha = t_\alpha(\bar{i}_\alpha), \qquad e_\beta = u_\beta(\bar{j}_\beta)$$

in the skeleton of I. Let $S_d, S_e \subseteq I$ be the unions of the skeleton supports of these presentations, and put $S = S_d \cup S_e$.

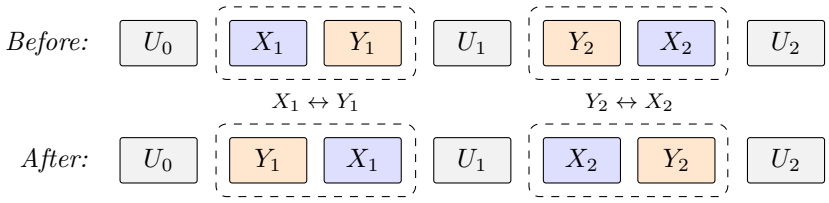
Suppose $(C_j)_{j \in J}$ is a family of pairwise disjoint convex subsets of the joint support order *S, indexed in increasing order: for $j < j'$ every point of C_j precedes every point of $C_{j'}$. Write*

$$X_j = C_j \cap S_d, \qquad Y_j = C_j \cap S_e.$$

Assume each region is the concatenation of two blocks, either $X_j < Y_j$ or $Y_j < X_j$, and that the blocks are exclusive*:*

$$X_j \cap S_e = \varnothing, \qquad Y_j \cap S_d = \varnothing.$$

In particular no skeleton coordinate used by both families lies in a transposed region. No assumption that X_j and Y_j are order-isomorphic is needed.



Schematic: X_j uses only coordinates of $\mathbf{d}$, Y_j only coordinates of $\mathbf{e}$, while the U_j denote untouched support (possibly shared). Each dashed region is convex in the joint support order*; only its two exclusive blocks are transposed. Internal block orders and term labels remain unchanged.*

Construct a new ordered support S' by replacing $X_j + Y_j$ with $Y_j + X_j$ (or conversely) for each j, retaining the internal order of every block and the order of all other support positions. Regard the same support coordinates as labeled positions in S': each occurrence of a coordinate in t_α or u_β retains its original assignment to that coordinate. Thus the first family still uses the X_j-coordinates and the second still uses the Y_j-coordinates; only their relative order changes*. Evaluate these same terms in an EM model over an order J containing S', obtaining tuples $\mathbf{d}'$ and $\mathbf{e}'$. Then*

$$\mathrm{atp}_{\mathrm{nat}}((\mathbf{d},\mathbf{e}); \mathrm{EM}_\tau(I,\Phi)) = \mathrm{atp}_{\mathrm{nat}}((\mathbf{d}',\mathbf{e}'); \mathrm{EM}_\tau(J,\Phi)),$$

The new tuples need not live in the original M; the statement compares their natural diagrams in (possibly different) EM presentations.

Proof sketch. Suppose the original and transposed diagrams differ. Choose σ bounding the two term families and supports, and χ sufficiently large with $\chi^\sigma = \chi$ for the orbit bound in source Lemma 4.5. Repeat the exclusive blocks along an auxiliary order L, with orientations chosen so that the diagram of a first-family tuple d_i with a second-family tuple e_j is the original diagram when $i < j$, and the transposed diagram when $i > j$.

Choose L with at most χ divider positions and more than χ branch positions realizing distinct cuts over the dividers. Explicitly, let ϵ be the least infinite cardinal with $2^\epsilon > \chi$ and use the lexicographically ordered binary tree $2^{<\epsilon}$, with a divider between each pair of successor subtrees and a point for each branch. Fix the tuples e_j at divider positions, together with the shared support, as parameters (at most χ of them). For distinct branch indices i, i', a divider j separates them, and the two diagrams of (d_i, e_j) and $(d_{i'}, e_j)$ differ. Hence the d_i realize more than χ automorphism orbits over these parameters. Embed the construction into a sufficiently large categoricity EM presentation, contradicting Lemma 4.5. □

Remark 3.3 (No-order-property interpretation of Lemma 4.6)**.** Interpretive remark, not a separately stated theorem of the source. *If swapping exclusive blocks changed a joint natural diagram from p to $q \neq p$, the repetition construction in Lemma 4.6 would yield, for arbitrarily large linear orders L, tuples $(\mathbf{d}_i)_{i\in L}$ and $(\mathbf{e}_j)_{j\in L}$ satisfying*

$$D(\mathbf{d}_i, \mathbf{e}_j) = \begin{cases} p, & i < j, \\ q, & i > j. \end{cases}$$

Thus a failure of block symmetry would witness an order property for canonical diagrams. *Lemma 4.5 rules this out by bounding automorphism orbits over a small parameter set. The conceptual progression is*

bounded automorphism orbits $\Longrightarrow$ *no such diagram order pattern* $\Longrightarrow$ *disjoint-block symmetry.*

This resembles the classical stability argument turning indiscernible sequences into indiscernible sets. It does not say that all order information on an EM skeleton is invisible in the reduct: only the specified transpositions of disjoint exclusive support blocks are covered.

Ordinary first-order no-order-property is not automatically sufficient: the term families and diagrams may have arbitrary set-sized length, bounded by σ*, and amalgamation has not yet been established. A cardinal-parameterized no-canonical-diagram-order-property principle might subsume Lemma 4.6, but the exact formulation and its relation to Lemma 4.5 remain to be checked.*

Problem for later. *Formulate a parameterized* no-order-property for natural arrow types *which implies disjoint block transposition. Compare it first to the first-order stable-theory argument that indiscernible sequences are indiscernible sets, and then ask whether an appropriate stability or superstability hypothesis for AECs with a monster model implies that principle. The present proof uses unbounded categoricity to obtain the uniform orbit bounds, so it does not immediately provide such a result.*

3.2 Locality and compactness for placed comparisons

Natural arrow types over parameter sets. Fix a well-ordered EM presentation $A = \mathrm{EM}_\tau(I, \Phi)$ and a tuple-length bound σ. For any set $X \subseteq A$, let

$$S^\sigma_{\mathrm{atp}_{\mathrm{nat}}}(X; A) := \{\mathrm{atp}_{\mathrm{nat}}((\mathbf{d}, \bar{x}); \mathrm{EM}_\tau(J, \Phi)) : I \hookrightarrow J,\ \mathbf{d} \in \mathrm{EM}_\tau(J, \Phi)^\sigma\},$$

where $\bar{x}$ is an enumeration of X, transported along the order-induced embedding. The comparison identifies the parameter coordinates according to their fixed names; its equality does not depend on the choice of enumeration. For $Y \subseteq X$, restriction of the parameter list gives a well-defined map

$$\restriction Y : S^\sigma_{\mathrm{atp}_{\mathrm{nat}}}(X; A) \longrightarrow S^\sigma_{\mathrm{atp}_{\mathrm{nat}}}(Y; A).$$

We suppress the ambient presentation A when it is fixed, writing $S^\sigma_{\mathrm{atp}_{\mathrm{nat}}}(X)$. The same construction works for *any subset* X of an EM base, whether or not X is the universe of a strong submodel. It does not automatically define a natural arrow type over an arbitrary non-EM base.

The source's Proposition 4.8 gives a definable nondecreasing cardinal function $g(\sigma)$ with the following *separation property*: two distinct placed natural arrow types over A are distinguished by their restrictions to some $X \subseteq A$ with $|X| \leq g(\sigma)$. This is tameness-like locality for the *placed comparison* invariant, not a general tameness statement for ordinary Galois types.

Lemma 3.4 (Unique compactness for natural arrow types over an EM base; consequence of [2, Propositions 4.8–4.9])**.** *Let* I *be an infinite well-order,* $A = \mathrm{EM}_\tau(I, \Phi)$*, and let* σ *be infinite. Choose an infinite cardinal* χ *satisfying*

$$\chi > \lambda_0 + \sigma, \qquad \chi^\sigma = \chi, \qquad \chi \geq g(\sigma).$$

For each $X \in [A]^{\leq \chi}$*, suppose we have specified a* possible *type*

$$p_X \in S^\sigma_{\mathrm{atp}_{\mathrm{nat}}}(X; A)$$

such that

$$X \subseteq Y, \quad |Y| \le \chi \quad \Longrightarrow \quad p_Y \restriction X = p_X.$$

Then there exists exactly one

$$p \in S^{\sigma}_{\mathrm{atp}_{\mathrm{nat}}}(A; A)$$

satisfying $p \restriction X = p_X$ *for all* $X \in [A]^{\le \chi}$. *Uniqueness is of the* type, *not of a realizing tuple or an order extension.*

Proof. Choose an enumeration for every parameter set, with the identifications made explicit when taking restrictions. Each p_X is, by definition, a possible joint canonical diagram of an added tuple with the parameters X. Its parameter restriction is the natural diagram of X, and the assumed coherence is exactly the compatibility hypothesis of source Proposition 4.9. That proposition produces a placement of $\mathbf{d}$ over I realizing all the prescribed diagrams, hence a type $p \in S^{\sigma}_{\mathrm{atp}_{\mathrm{nat}}}(A; A)$ with $p \restriction X = p_X$.

If p, p' are two such types, source Proposition 4.8 finds, whenever $p \neq p'$, a subset $X \subseteq A$ of size at most $g(\sigma) \le \chi$ on which they differ. Both restrictions equal p_X, contradicting the choice of X. □

Remark 3.5 (The quantifier over parameter subsets)**.** *The statement concerns* $X \in [A]^{\le \chi}$. *Asking for* p_X *for* all *subsets* $X \subseteq A$, *including* $X = A$, *would already supply the desired conclusion as* p_A. *If* $|A| \le \chi$, *the theorem is correspondingly tautological; its interest is when* $|A| > \chi$. *The hypothesis that each* p_X *is* possible *is essential.*

If $A_0 = \mathrm{EM}_\tau(I_0, \Phi) \le_{\mathbf{K}} A$ *for an infinite sub-well-order* $I_0 \subseteq I$, *the same lemma applies directly with* A_0 *as the base. This covers EM submodels. For an arbitrary subset* $Y \subseteq A$, *the notation* $S^{\sigma}_{\mathrm{atp}_{\mathrm{nat}}}(Y; A)$ *is still meaningful, but the stated theorem is over the* whole EM base A. *A corresponding compactness theorem based only on prescriptions on* $[Y]^{\le \chi}$ *would require adapting the support-profile argument of source Proposition 4.9 and the uniform testing bound of Proposition 4.8 to parameters restricted to* Y. *We do not assert that extension here. Likewise, arbitrary strong submodels of* A *need not be EM submodels with the induced presentation.*

Proposition 3.6 (Gluing types along a χ-covering chain)**.** *Keep the hypotheses on* A, σ, χ *from Lemma 3.4. Suppose* $A = \bigcup_{i<\delta} A_i$ *for an increasing family of parameter subsets (not necessarily submodels), and each* $X \in [A]^{\le \chi}$ *is contained in some* A_i. *If*

$$p_i \in S^{\sigma}_{\mathrm{atp}_{\mathrm{nat}}}(A_i; A) \qquad (i < \delta)$$

satisfy $p_j \restriction A_i = p_i$ *whenever* $i < j$, *then there is a unique* $p \in S^{\sigma}_{\mathrm{atp}_{\mathrm{nat}}}(A; A)$ *extending all the* p_i. *In particular, the covering assumption holds whenever* $\mathrm{cf}(\delta) > \chi$.

Proof. For each $X \in [A]^{\le \chi}$, choose i with $X \subseteq A_i$ and put $p_X = p_i \restriction X$. This is independent of the chosen i: any two indices have a common later index, and the chain is coherent. The p_X are possible and satisfy the compatibility conditions of Lemma 3.4, which gives existence and uniqueness of p. For the last assertion, if $|X| \le \chi$, choose for each $x \in X$ an index i_x with $x \in A_{i_x}$. Since $\mathrm{cf}(\delta) > \chi$, the supremum of those indices is below δ, so X lies in some A_i. □

Remark 3.7 (The countable-union problem remains open here)**.** *Suppose* $A = \bigcup_{n<\omega} A_n$ *and* $p_n \in S^{\sigma}_{\mathrm{atp}_{\mathrm{nat}}}(A_n; A)$ *are coherent under restriction. The assumption in Proposition 3.6 usually fails: one can choose elements from cofinally many successive stages to obtain a countable set* $X \subseteq A$ *which is contained in no* A_n. *Since* χ *is infinite, the* p_n *alone do not assign a diagram to this* X. *Thus Lemma 3.4 does not prove that the* p_n *have a unique common extension over* A.

One might seek a stronger countable-chain continuity *principle for placed natural arrow types. Its existence and uniqueness parts are distinct questions: even if a global extension exists, the uniform testing bound $g(\sigma)$ only separates different global types over a parameter set of size at most $g(\sigma)$, which need not lie in one A_n. For comparison, the source's Lemma 4.10 handles* bounded diagrams of an actual tuple in an ambient strong extension *and uses additional closed-cardinal hypotheses; it does not state arbitrary countable-chain continuity for abstract type prescriptions.*

If p_n are instead all types over the same base A, *there may be a different issue: their* tuple lengths *could be growing. This is distinct from the chain of parameter sets considered here; see the next remark. If the base and tuple length are both fixed, an increasing sequence in the sense of restriction is simply a constant type.*

Remark 3.8 (Countable unions in the tuple coordinates)**.** *Fix the same EM base A and ordinals γ_n with $\gamma_n \le \gamma_{n+1}$, $\gamma = \sup_{n<\omega} \gamma_n$. Suppose*

$$p_n \in S^{\gamma_n}_{\mathrm{atp}_{\mathrm{nat}}}(A; A), \qquad p_{n+1} \restriction \gamma_n = p_n,$$

where the restriction now discards tuple coordinates*, not parameters. Does there exist a unique $p \in S^{\gamma}_{\mathrm{atp}_{\mathrm{nat}}}(A; A)$ with $p \restriction \gamma_n = p_n$?*

This is a different continuity question. Our compact-placement Lemma 3.4 works at a fixed *upper bound on the entire added tuple length. To use it here, one would first need, for every $X \in [A]^{\le \chi}$, a* possible *joint diagram q_X for the entire γ-tuple with parameters X, whose restrictions agree with $p_n \restriction X$ for all n and whose parameter restrictions are coherent. If such q_X are supplied and $\gamma \le \sigma$ with the stated conditions on χ, the lemma gives a unique resulting* full *type. But existence of these q_X is not a consequence proved here of the coherence of the p_n alone.*

Even assuming existence, to show uniqueness from the p_n alone *one would need a continuity principle saying that two natural arrow types of a long tuple which agree on every initial γ_n-subtuple are equal. The uniform testing theorem is about* parameter subsets *of bounded size, not continuity in the number of distinguished tuple coordinates. This is precisely the distinction between a* tameness-like *principle (small parameter sets detect unequal types) and a* type-shortness-like *principle (small coordinate subtuples detect unequal long types). The source proves the former for placed comparisons, but the latter is not a stated consequence.*

The countable exhaustion in the proof of source Theorem 4.11 uses actual compatible enumerations *of strong submodels and their already determined bounded diagrams. Its construction extends realizations stage by stage. It does not assert this unrestricted inverse-limit property for arbitrary abstract types of growing tuple length. This distinction is worth revisiting when auditing that construction.*

Remark 3.9 (Scope of compactness and uniqueness)**.** *This is a compactness-and-separation principle for* placed canonical diagrams over well-ordered EM presentations*, not a compactness theorem for arbitrary Galois types. The possibility hypothesis on each q_X is essential; coherence alone does not assert that an arbitrary syntactic prescription is a possible joint diagram. The bound $g(\sigma)$ is independent of $|I|$. In Lemma 4.10 the source applies this principle at cofinally many $\chi < \mu$ to lift all bounded diagrams of an actual tuple over a base of size μ.*

Remark 3.10 (The role of cofinality in the next step)**.** *The compactness and uniform-testing propositions are formulated for arbitrary infinite well-ordered skeletons. In the source, Lemma 4.10 applies them at strong-limit cardinals μ of countable cofinality that are strictly closed below themselves under the relevant threshold functions t and g. The proof of that lemma uses cofinally many $\chi < \mu$ with $\chi^{\sigma} = \chi$ (available from the strong-limit assumption), whereas countable cofinality is central*

to the later countable exhaustion used for amalgamation in Theorem 4.11. The weaker cofinality requirements that might suffice for Lemma 4.10 alone have not been established as a separate source theorem.

Theorem 3.11 (Amalgamation at selected levels; [2, Theorem 4.11])**.** *There is an unbounded class of strong-limit cardinals μ of countable cofinality such that:*

(1) $\mathbf{K}$ *is categorical in μ, and two μ-sized strong extensions of a μ-model admit a strong amalgam of size μ.*

(2) For bounded tuples $\bar{d}_i \in B_i$ over a common μ-sized base A, an amalgam identifying the tuples exists exactly when their relative diagrams agree on all $X \subseteq A$ of size at most $g(\sigma)$, for the relevant tuple-length bound σ.

(3) There are at most μ singleton types over each μ-model, where "type" is identification in an amalgam over the model.

Proposition 3.12 (Placed natural arrow types are Galois types at closed levels)**.** *Let μ be a cardinal to which source Lemma 4.10 and Theorem 4.11 apply. Suppose $A \leq_{\mathbf{K}} B_0, B_1$ are all of cardinality μ and $\mathbf{d}_i \in B_i^{\gamma}$ for the same $\gamma \leq \sigma < \mu$. Fix an isomorphism $A \cong \mathrm{EM}_{\tau}(I, \Phi)$ for a well-order I of size μ. Let $p_i \in S^{\gamma}_{\mathrm{atp}_{\mathrm{nat}}}(A; A)$ be the* unique placed lift *of the bounded diagrams*

$$\mathrm{atp}((\mathbf{d}_i, \bar{x}); B_i), \qquad X \subseteq A, \quad |X| < \mu,$$

where $\bar{x}$ enumerates X, and the arrow type is computed at a stabilization level below μ appropriate to the tuple length. Then the following are equivalent:

(i) *$p_0 = p_1$;*

(ii) *the bounded diagrams of $(\mathbf{d}_0, X; B_0)$ and $(\mathbf{d}_1, X; B_1)$ agree for every $X \subseteq A$ with $|X| \leq g(\sigma)$;*

(iii) *there exists a strong amalgam of B_0, B_1 over A identifying $\mathbf{d}_0, \mathbf{d}_1$;*

(iv) *$\mathrm{gtp}(\mathbf{d}_0/A; B_0) = \mathrm{gtp}(\mathbf{d}_1/A; B_1)$, with Galois types defined by unrestricted transitive closure over A.*

In particular, at such levels the equality relation on placed natural arrow types of tuples in μ-sized extensions coincides with *equality of ordinary Galois types over the μ-base.*

Proof. The existence and uniqueness of p_i is source Lemma 4.10. Source Proposition 4.8 shows that two placed lifts agree if and only if their restrictions to parameter sets of size at most $g(\sigma)$ agree; this gives (i)⇔(ii). The equivalence of (ii) and (iii) is exactly source Theorem 4.11(2).

An amalgam identifying the tuples immediately gives equality of Galois types. Conversely, a chain witnessing equality of Galois types *over the fixed base A* cannot pass through models smaller than μ, since every intermediate model contains a strong copy of A. Downward Löwenheim–Skolem and coherence shrink each direct comparison to a comparison among μ-sized strong extensions of A (and μ-sized amalgams). The amalgamation property for such models, asserted in Theorem 4.11(1), makes direct equivalence over A transitive. Consequently the entire finite chain collapses to a single identifying μ-amalgam. This proves (iv)⇒(iii). □

Remark 3.13 (Not simply a new name for Galois types)**.** *The preceding proposition provides a* Galois-type characterization after amalgamation has been proved*, at the special closed levels of Theorem 4.11. Before that theorem, stabilized arrow types and their placed lifts are available without assuming amalgamation, and they supply the comparison machinery used to* establish *amalgamation. This is the distinctive role of the construction in the manuscript. On arbitrary smaller parameter subsets of A, the placed invariant is defined, but it need not already coincide with ordinary Galois types over that subset.*

Both eventual reversal (source Theorem 3.11) and stabilization (source Proposition 4.2) use unbounded categoricity essentially in their stated proofs. In an AEC with amalgamation and a monster model, Galois types already admit an automorphism-orbit description, whereas the directed strong-map comparison between arbitrary models of size κ need not be symmetric without additional universality or homogeneity. It is an open question for these notes whether an appropriate version of the arrow-type machinery can be useful in superstable AECs without the unbounded categoricity hypothesis; no such general theorem is being claimed here.

Remark 3.14 (Scope of amalgamation)**.** *This is amalgamation at selected cardinals, not an assertion of eventual amalgamation throughout a tail. The first conclusion is distinct from the uniform tameness and compactness of placed diagrams established earlier.*

4 Functor categories and working schemes

4.1 Functor categories of AEC models

We replace the graph-based language of systems in [2, Section 5.1] with the category of diagrams of AEC models. This is a reformulation in our notes, not terminology used explicitly in the source.

Definition 4.1 (The diagram category)**.** *Regard* $\mathbf{K}$ *as the category whose objects are models in the AEC and whose morphisms are* $\mathbf{K}$*-embeddings. Fix a* finitely generated small category $\mathcal{I}$*, meaning a category generated by a finite set of objects and arrows under identities and composition. Set*

$$\mathbf{K}^{\mathcal{I}} := \mathrm{Fun}(\mathcal{I}, \mathbf{K}).$$

An object is a functor $D : \mathcal{I} \to \mathbf{K}$*: it assigns a model* $D(v)$ *to each object* v *and a strong embedding* $D(\alpha) : D(v) \to D(w)$ *to each arrow* $\alpha : v \to w$*. A morphism* $\eta : D \Rightarrow E$ *is a natural transformation whose components* $\eta_v : D(v) \to E(v)$ *are strong embeddings:*

$$E(\alpha) \circ \eta_v = \eta_w \circ D(\alpha) \qquad (\alpha : v \to w).$$

An isomorphism is a natural isomorphism, equivalently a commuting family of component isomorphisms.

Remark 4.2 (From graphs to free categories)**.** *For a directed multigraph* G*, let* $\mathcal{F}(G)$ *be its* free category*. The morphisms are finite directed paths (including empty identity paths), and composition concatenates paths. Every choice of* $\mathbf{K}$*-models at the vertices and strong embeddings along the edges extends uniquely to a functor* $\mathcal{F}(G) \to \mathbf{K}$*. Graph-system embeddings become precisely the natural transformations. No commutativity is imposed on distinct paths. This is the* free-category *construction, not the nerve of a category.*

Every category generated by a finite graph is a quotient of its free category by a congruence on paths. Consequently a finitely generated category *may have infinitely many morphisms but has at most*

countably many. This encompasses the free category on a finite graph (which can have infinite path sets if there are directed cycles), as well as every finite category, including the finite posets used for commuting cubes.

We do not *assume that $\mathcal{I}$ is finitely presented: relations among paths may be infinite in number. A category with relations forces its diagrams to satisfy the corresponding commutativities; the source's arbitrary finite graph systems impose none. When applying the manuscript's working package to an additional category with relations, we must still check its particular presentation and arrow axioms.*

Definition 4.3 (The auxiliary category of structural embeddings)**.** *Let $\mathbf{K}^{*\tau}$ have exactly the models of $\mathbf{K}$ as objects, but take*

$$\mathrm{Hom}_{\mathbf{K}^{*\tau}}(M,N) := \mathrm{Emb}_{\tau(\mathbf{K})}(M,N),$$

the structural *embeddings, which need not have strong image. Then $\mathbf{K}$ is a subcategory of $\mathbf{K}^{*\tau}$, and we have a faithful inclusion*

$$\mathrm{Fun}(\mathcal{I},\mathbf{K}) \longrightarrow \mathrm{Fun}(\mathcal{I},\mathbf{K}^{*\tau}).$$

The source objects on the left have strong internal maps, whereas diagrams on the right have merely structural internal maps. This is not a full inclusion*, because the right-hand category also allows natural transformations whose components are not strong.*

Define $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})$ *to have all objects of* $\mathrm{Fun}(\mathcal{I},\mathbf{K}^{*\tau})$ *but only natural transformations $\eta : D \Rightarrow E$ whose components $\eta_v : D(v) \to E(v)$ are* strong*. Composition and identities preserve componentwise strongness. The category* $\mathrm{Fun}(\mathcal{I},\mathbf{K})$ *is a full subcategory of* $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})$ *on the diagrams whose* internal *arrows are all strong. Thus*

$$\mathrm{Fun}(\mathcal{I},\mathbf{K}) \;\subseteq\; \mathrm{Diag}_{\mathcal{I}}(\mathbf{K}) \;\subseteq\; \mathrm{Fun}(\mathcal{I},\mathbf{K}^{*\tau}),$$

where only the first inclusion is full.

Remark 4.4 (What the manuscript requires of internal maps)**.** *In source Section 5.1 the maps* between vertices inside a system *are called* specified maps*. The arrows* between systems*, by contrast, are explicitly componentwise strong, possibly with additional conditions. The general wording does not explicitly say that every specified internal map is strong. If those maps are τ-embeddings,* $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})$ *is a faithful categorical reformulation.*

If some permitted internal map is not even a structural embedding, then neither functor category above describes that generality without further modification. The later marked-increment and finite systems constructions use specified strong internal maps, so they fit $\mathrm{Fun}(\mathcal{I},\mathbf{K})$*. We retain the broader structural-embedding case here as a clarification of the general working-package setup rather than asserting that the manuscript needs it in every application.*

Definition 4.5 (Many-sorted coding and strong subfunctors)**.** *Use one sort S_v per object $v \in \mathcal{I}$, a copy of $\tau(\mathbf{K})$ on that sort, and a function symbol $F_\alpha : S_v \to S_w$ for each $\alpha : v \to w$. Functoriality means*

$$F_{\mathrm{id}_v} = \mathrm{id}_{S_v}, \qquad F_{\beta\alpha} = F_\beta F_\alpha.$$

The sorted structures coding functors $D : \mathcal{I} \to \mathbf{K}^{*\tau}$ have carriers $S_v^D = D(v)$ and maps $F_\alpha^D = D(\alpha)$, which are τ-embeddings. For sorted substructures $D \subseteq E$, set

$$D \le_{\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})} E \quad\Longleftrightarrow\quad D(v) \le_{\mathbf{K}} E(v) \quad \textit{for every } v.$$

Thus the strong embeddings of the sorted structures are precisely the componentwise strong *natural transformations. The class of diagrams with all internal arrows strong recovers* $\mathrm{Fun}(\mathcal{I},\mathbf{K})$ *as a full subcategory.*

Proposition 4.6 (The strong diagram categories are AECs)**.** *For any fixed small category $\mathcal{I}$, the sorted diagrams represented by* $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})$*, with componentwise strong-subfunctor relation, form an AEC with*

$$\mathrm{LS}(\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})) \leq \max\{\mathrm{LS}(\mathbf{K}), |\mathrm{Mor}(\mathcal{I})|, \aleph_0\}.$$

The full subcategory $\mathrm{Fun}(\mathcal{I}, \mathbf{K})$*, where all* internal *arrows are also strong, likewise forms an AEC. In particular, for any finitely generated shape $\mathcal{I}$, both LS numbers are at most* $\mathrm{LS}(\mathbf{K})$*.*

Proof. Isomorphism invariance and coherence follow componentwise. For an increasing chain of strong subfunctors, take unions of vertex models and diagram maps in each sort. Each component union is strong by the AEC chain axioms, and the maps on the union remain structural embeddings and preserve functoriality. In the full subcategory of diagrams with strong internal maps, they remain strong by the same chain axiom.

For downward Löwenheim–Skolem, fix an infinite κ at least the displayed bound and a set of at most κ sorted elements of a diagram $D : \mathcal{I} \to \mathbf{K}^{*\tau}$. Iteratively construct strong submodels $M_v^n \leq_{\mathbf{K}} D(v)$ of size at most κ for every vertex v, enlarging them at stage $n+1$ to contain the images $D(\alpha)[M_v^n]$ for every arrow $\alpha : v \to w$. There are at most κ such elements at each stage. For each v, take the countable union $M_v = \bigcup_{n<\omega} M_v^n$. Then $M_v \leq_{\mathbf{K}} D(v)$ and $D(\alpha)[M_v] \subseteq M_w$. The restricted maps are τ-embeddings, so these vertex models form a strong subfunctor. If the original $D(\alpha)$ were strong, the restrictions remain strong by coherence, proving the result for $\mathrm{Fun}(\mathcal{I}, \mathbf{K})$ as well. □

Definition 4.7 (Sorted tuples and named data)**.** *A sorted tuple of sort pattern $\mathbf{v} = (v_i)_{i<\gamma}$ in $D : \mathcal{I} \to \mathbf{K}^{*\tau}$ is a tuple $\mathbf{a} = (a_i)_{i<\gamma}$ with $a_i \in D(v_i)$. Thus sorting just fixes the sort of each coordinate, as in ordinary many-sorted model theory. Equal underlying elements occurring in distinct vertex sorts remain distinct tagged occurrences; functor-map symbols record their specified relationships. Distinguished lists are named sorted tuples.*

Definition 4.8 (Working schemes over the diagram category)**.** *The ambient category of* structural internal diagrams with strong external morphisms *is* $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})$*. A working scheme is most naturally regarded as a category $\mathcal{C}$ equipped with a* faithful forgetful functor

$$U : \mathcal{C} \longrightarrow \mathrm{Diag}_{\mathcal{I}}(\mathbf{K}).$$

Its objects are eligible diagrams, possibly equipped with named elements, distinguished submodels, or other additional data*. Its morphisms become componentwise strong natural transformations under U, subject to any further scheme-specific conditions.*

For an undecorated *working scheme, $\mathcal{C}$ may be identified with a not necessarily full subcategory of* $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})$*. When marked data are included, distinct objects of $\mathcal{C}$ can have the same underlying diagram $U(D)$ but different marks. Then $\mathcal{C}$ is* not literally *a subcategory of the category of unmarked diagrams. For example, a named-tuple scheme has objects $(D, \mathbf{a})$ and arrows $\eta : D \Rightarrow E$ satisfying $\eta(\mathbf{a}) = \mathbf{b}$. These are categories of structured* diagram objects*, not categories of subobjects of one fixed functor.*

When every internal arrow is strong, the underlying diagrams lie in the full subcategory $\mathrm{Fun}(\mathcal{I}, \mathbf{K})$*. The working package later imposes suitable strong-subdiagram closure, homogeneity and presentation properties. It does not assert that every such eligible working category is itself an AEC.*

Scope of the source. *The manuscript's general phrase* specified maps *does not force internal maps to be strong; our broader diagram category permits structural embeddings internally. We have not established that a later application requires strictly non-strong internal maps.*

4.2 Eligible levels, EM presentations, and comparisons

Definition 4.9 (Eligible levels and closure bounds)**.** *Fix an unbounded definable class $\mathcal{S}$ of infinite cardinals, the* eligible levels*. A functor $D : \mathcal{I} \to \mathbf{K}^{*\tau}$ is at level $\mu \in \mathcal{S}$ when every component $D(v)$ has cardinality μ. This is a designated part of the diagram AEC, not an assertion that all μ-sized objects of* $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})$ *have all vertex sorts of size μ. A sorted list is* bounded at μ *if its length is strictly less than μ.*

In source Convention 5.1, u is a placeholder for any specified cardinal-valued bound function*, not an additional model-theoretic object. For such u and a cardinal μ, say that μ is* strictly closed under u *when*

$$u(\theta) < \mu \quad \text{for every infinite cardinal } \theta < \mu.$$

Examples of bounds to which this convention applies are the earlier stabilization and testing functions $t(\theta)$ and $g(\theta)$, as well as the coding bound $b_F(\theta)$ of source Proposition 5.9. One may impose simultaneous strict closure under any specified set of such functions. This does not by itself imply $\mu \in \mathcal{S}$; eligible-level membership and the availability of cofinally many smaller eligible levels are additional questions (source Section 5.2).

Definition 4.10 (EM-style presentation of functor diagrams)**.** *Fix a set-sized collection of finite-arity term labels, each with an output sort $v \in \mathrm{Ob}(\mathcal{I})$. An* EM-style presentation template *F associates to each well-order L of eligible cardinality a diagram object*

$$F(L) : \mathcal{I} \longrightarrow \mathbf{K}^{*\tau}, \qquad F(L) \in \mathrm{Diag}_{\mathcal{I}}(\mathbf{K}).$$

Order embeddings between well-orders of eligible cardinalities are required to induce strong natural transformations between these diagrams. Every element of every component $F(L)(v)$ is represented by a label applied to a finite increasing tuple from the common indexing order L:

$$a = t(i_0, \dots, i_{n-1}), \qquad i_0 < \cdots < i_{n-1} \text{ in } L.$$

Equality, vertex relations and functions, and the maps $F(L)(\alpha)$ are all computed uniformly from the label data and finite order/equality patterns of the supporting positions. Different labeled expressions may represent the same sorted element.

This is a many-sorted EM construction with one external common skeleton *L. One can make it a literal EM blueprint by adjoining a skeleton sort and symbols for evaluating the labels into their respective vertex sorts. The working package separately requires order embeddings between well-orders of eligible size to induce* strong natural transformations *$F(L) \Rightarrow F(L')$ and requires every eligible diagram to be isomorphic to one of these presentations at its level. These stronger assertions are not automatic for every diagram AEC. For example, even when $\mathbf{K}$ is the AEC of pure sets, the strong-diagram AEC for the walking-arrow category $v \to w$ is not categorical in infinite cardinality μ among diagrams with both vertices of size μ: one can use a surjective map or a proper injection. The two diagrams are not naturally isomorphic.*

In the source, $F(L)$ is initially assumed to be an eligible diagram only for well-orders L with $|L| \in \mathcal{S}$; extending the presentation to padded or arbitrary orders is a later construction (source Lemma 5.4), not part of this initial definition.

Definition 4.11 (Diagram comparisons)**.** *For each sort pattern $\mathbf{v} = (v_i)_{i<\gamma}$, with γ bounded at the relevant levels, a* diagram comparison *is an equivalence relation $\equiv_D^{\mathbf{v}}$ on pairs $(\mathbf{a}; \mathcal{D})$ where $\mathcal{D} : \mathcal{I} \to \mathbf{K}^{*\tau}$ is an eligible functor and $\mathbf{a}$ is a sorted tuple of pattern $\mathbf{v}$. Write*

$$D(\mathbf{a}; \mathcal{D}) = D(\mathbf{b}; \mathcal{E}) \quad \Longleftrightarrow \quad (\mathbf{a}; \mathcal{D}) \equiv_D^{\mathbf{v}} (\mathbf{b}; \mathcal{E}).$$

In source Definition 5.2 these equivalence relations are presented as data. In our alternate definition below, they will instead be recovered as lifted automorphism types. Equivalence and the same-level orbit characterization remain substantive hypotheses, not consequences of the ambient diagram category; the appendix proves the comparison.

Definition 4.12 (Possible joint diagram)**.** *For a bounded sorted tuple* $\mathbf{a}$ *in an eligible functor* $\mathcal{D}$*, a diagram* $q(\mathbf{d}, \mathbf{a})$ *is* possible over $\mathbf{a}$ *if there exist an eligible functor* $\mathcal{E}$ *and a sorted tuple* $(\mathbf{d}', \mathbf{a}')$ *of bounded total length such that*

$$D(\mathbf{d}', \mathbf{a}'; \mathcal{E}) = q, \qquad D(\mathbf{a}'; \mathcal{E}) = D(\mathbf{a}; \mathcal{D}).$$

Possibility does not initially imply realizability over the actual tuple $\mathbf{a}$ *in* $\mathcal{D}$*; that is a homogeneity/saturation assertion of the working package.*

4.3 Working schemes via generated AECs and lifted types

We now use Boney's alternate presentation as our definition. The source's four-clause package is restated in Appendix A, where we prove the comparison, with its precise scope restrictions. This is a reorganization of the package, not a new verification of any of the manuscript's applications.

Definition 4.13 (Concrete coding convention)**.** *Fix a nonempty finitely generated shape* $\mathcal{I}$*, an AEC* $\mathbf{K}$*, and an unbounded definable class* $\mathcal{S}$ *of infinite cardinals with minimum* $\mu_0 \geq \mathrm{LS}(\mathbf{K})$*. Use a fixed finitary many-sorted language* τ^{gen}*, of size at most* μ_0*, coding the vertex structures, specified internal maps, and all fixed named data. Its universe is the disjoint union of the vertex occurrences; no indexing skeleton is added to this universe. Structural embeddings and isomorphisms always refer to this entire code, including the names. Write* $\|X\|$ *for its total cardinality and* $|X|$ *for its sorted universe.*

An AEC $\mathbf{K}^{\mathrm{gen}}$ over $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K}_{\geq \mu_0})$ *means an AEC of these codes, with no models below* μ_0*, together with the concrete faithful forgetful functor* U *which drops the named data. Its strong embeddings have underlying structural embeddings with strong vertex maps; the converse is not required. Sorted universes and maps are retained by* U*, and directed colimits are computed on them, so* U *preserves directed colimits. We do not require* U *to be full or injective on objects.*

In this diagram version, the internal maps are structural embeddings. For the source's more general specified maps, replace the target by the category of finite systems of $\mathbf{K}$*-models with arbitrary specified functions and componentwise strong commuting embeddings. The same sorted-code convention and the arguments below apply to that version. This change of ambient category is necessary for a claim covering that full generality of the source.*

Definition 4.14 (Structural data and lifted automorphism comparison)**.** *In the convention above, the data are:*

(D1) *An AEC* $\mathbf{K}^{\mathrm{gen}}$ *over the diagram category with* $\mathrm{LS}(\mathbf{K}^{\mathrm{gen}}) = \mu_0$*.*

(D2) *A set-sized finite-support term recipe* Ψ *with values denoted* $F(L) = \mathrm{EM}_{\tau^{\mathrm{gen}}}(L, \Psi)$*. Here this notation denotes the* system code*, not a structure with an additional named skeleton. Elements have sorted labels on finite increasing supports; equality and all symbols of the code are evaluated uniformly from labels and finite order/equality patterns. This includes fixed named data and compatibility with substitution of positions. We require* $F(\alpha) \in \mathbf{K}^{\mathrm{gen}}$ *with every vertex of size* μ *whenever* $\mu \leq \alpha < \mu^+$ *and* $\mu \in \mathcal{S}$*. No membership assertion for arbitrary linear orders is included.*

(D3) *Let $\mathcal{C}_\mu$ be the structured-isomorphism closure of these $F(\alpha)$, and put $\mathcal{C} = \bigcup_{\mu\in\mathcal{S}} \mathcal{C}_\mu$. Specify*

$$H(X,Y) \subseteq \mathrm{Emb}_{\mathbf{K}^{\mathrm{gen}}}(X,Y) \qquad (X,Y \in \mathcal{C}).$$

Write H_μ for the same-level arrows, and require equality of both objects and arrows

$$(\mathcal{C}_{\mu_0}, H_{\mu_0}) = \mathbf{K}^{\mathrm{gen}}_{\mu_0}.$$

For tuples of one fixed sorted index pattern, bounded in both containing objects, define the relation E as follows: $E((\mathbf{a};X),(\mathbf{b};Y))$ holds if there exist an eligible Z, $f \in H(X,Z)$, $g \in H(Y,Z)$, and $h \in \mathrm{Aut}(Z)$ such that $hf(\mathbf{a}) = g(\mathbf{b})$. Once E is an equivalence relation, its classes are denoted $\mathrm{tp}_{\mathrm{la}}(\mathbf{a};X)$. All witnesses here are eligible *and all maps belong to H; arbitrary $\mathbf{K}^{\mathrm{gen}}$-amalgams are not being used.*

Definition 4.15 (Working scheme: alternate axioms)**.** *The preceding data form a working scheme if the following hold.*

(A1) ***Equivalence.*** *For every bounded sorted index pattern, E is an equivalence relation across the eligible objects large enough for that pattern. In particular, transitivity is a requirement, not an automatic property of common-target comparisons.*

(A2) ***Presentation arrows.*** *Every order embedding between well-orders of eligible, possibly different, cardinalities induces an arrow in H.*

(A3) ***Structural and fixed-level closure.*** *H contains all structured isomorphisms and is closed under composition. If $X \subseteq Y \subseteq Z$ are eligible and both inclusions into Z belong to H, then $X \hookrightarrow Y$ belongs to H. For each μ, $(\mathcal{C}_\mu, H_\mu)$ is an AEC in μ: the induced strong-substructure relation is coherent, invariant under isomorphism, and satisfies chain unions and smoothness at that size. Explicitly, for a nonempty increasing chain of inclusion arrows of length $0 < \delta < \mu^+$, its union X lies in $\mathcal{C}_\mu$, and every stage includes into X by an arrow. If the stages include by arrows into* any eligible *Y, their union includes into Y by an arrow as well. The larger-target clause is part of the axiom, not just same-level smoothness.*

(A4) ***Same-level orbit characterization.*** *For $X, Y \in \mathcal{C}_\mu$ and tuples of the same sorted pattern of length strictly below μ,*

$$\mathrm{tp}_{\mathrm{la}}(\mathbf{a};X) = \mathrm{tp}_{\mathrm{la}}(\mathbf{b};Y) \quad\Longleftrightarrow\quad \exists h : X \cong Y \ (h(\mathbf{a}) = \mathbf{b}).$$

The isomorphism is of the whole structured system. Include the empty tuple.

(A5) ***Countable exhaustion.*** *Let $X \in \mathcal{C}_\mu$, $Y \in \mathcal{C}_\nu$, and let $p : X \to Y$ be a structural embedding of codes, not initially assumed strong or in H. Suppose there are increasing subsets $A_n \subseteq |X|$, $n < \omega$, of size $< \mu$, with union $|X|$. For every vertex v, suppose also that*

$$B^v_n \leq_{\mathbf{K}} B^v_{n+1} \leq_{\mathbf{K}} X(v), \qquad |B^v_n| < \mu, \qquad \bigcup_{n<\omega} B^v_n = X(v), \qquad |B^v_n| \subseteq A_n.$$

Choose a sorted enumeration $\mathbf{a}_n$ of A_n and use its image enumeration $p(\mathbf{a}_n)$, with coordinate identifications consistent under inclusion. If, for every n and v,

$$\mathrm{tp}_{\mathrm{la}}(\mathbf{a}_n;X) = \mathrm{tp}_{\mathrm{la}}(p(\mathbf{a}_n);Y), \qquad p[B^v_n] \leq_{\mathbf{K}} Y(v),$$

then $p \in H(X,Y)$. The sets A_n need not be substructures. The B^v_n are additional witnesses contained in those sets; they need not equal $A_n \cap X(v)$. Any extra geometric arrow condition must be verified when establishing this axiom for a particular scheme.

For comparison with the main regime of source Convention 5.1, take eligible levels to be strong limits of countable cofinality. We keep the countable exhaustion formulation explicit. At preliminary levels of other cofinalities, equivalence with a more broadly indexed exhaustion axiom is not asserted.

Remark 4.16 (Which clauses of the alternate draft became consequences?)**.** *The draft separately lists mixed-arrow factorization, categoricity, term naturality, and realization over actual tuples. We retain all four as proved consequences below and in the appendix, rather than repeat them as independent axioms. Their deletion from the axiom list does not weaken the package. This also strengthens the draft's same-chart naturality statement to the source's cross-level formulation.*

Proposition 4.17 (Derived package)**.** *Under the alternate axioms:*

(i) *There are common eligible targets. Lifted types include atomic information, respect restrictions and reindexing, and are preserved by every H-arrow. There is one empty type, so $\mathcal{C}_\mu$ is categorical as a class of* systems *at each eligible μ.*

(ii) *Mixed downward closure holds: if $\nu \leq \mu$ are eligible, every set of $\leq \nu$ sorted occurrences in $X \in \mathcal{C}_\mu$ lies in an H-subobject in $\mathcal{C}_\nu$, retaining all fixed names.*

(iii) *For each fixed sorted arity θ, the number of lifted types has a set bound independent of level. A possible joint type is realized over an actual tuple at a receiving level whenever the joint length is bounded in both the witness and receiver.*

(iv) *Term tuples with the same labels and the same* joint *order/equality pattern of supports have equal lifted types, including across eligible levels.*

Proof. See Lemmas A.3–A.5 in the appendix. □

Proposition 4.18 (Mixed arrows from the family of level arrows)**.** *Let $X \in \mathcal{C}_\mu$, $Y \in \mathcal{C}_\nu$, $\mu < \nu$. Every $f \in H(X, Y)$ factors as jf_0, where $f_0 \in H_\mu(X, Y_0)$ and $j : Y_0 \to Y$ is, up to chart isomorphisms, induced by inclusion of a suborder of size μ in a chart for Y. Conversely, every such composite is in $H(X, Y)$. Thus the* family $(H_\mu)_{\mu \in \mathcal{S}}$ *and the presentation arrows determine all mixed arrows of a working scheme.*

Proof. Choose a chart for Y and supports for all occurrences in $f[X]$. Their union has size at most μ; enlarge it to a suborder J of size μ. Its presentation has image Y_0 containing $f[X]$, and its inclusion j is an arrow. The inclusion $f[X] \hookrightarrow Y$ is an arrow by isomorphism invariance. Coherence gives $f[X] \hookrightarrow Y_0$ in H, hence $f_0 \in H_\mu$. The converse follows by composition. Fixed names are preserved by the presentation arrows. This argument proves the factorization inside an existing package; declaring this formula for arbitrary level data would still require checking coherence and composition. □

Remark 4.19 (The small-level completion and its remaining fullness question)**.** *The standard upward-completion theorem for an AEC in μ_0 supplies*

$$\mathbf{K}^{\mathrm{gen}} = (\mathcal{C}_{\mu_0}, H_{\mu_0})^{\uparrow}, \qquad \mathrm{LS}(\mathbf{K}^{\mathrm{gen}}) = \mu_0.$$

It recovers both objects and arrows *at μ_0; see Appendix A.3. Earlier versions of these notes incorrectly left all the AEC axioms of this completion as outstanding. The actual unresolved issue here is higher-level fullness:*

$$H(X, Y) \stackrel{?}{=} \mathrm{Emb}_{\mathbf{K}^{\mathrm{gen}}}(X, Y) \quad \textit{for eligible } X, Y.$$

Every designated arrow has a coherent small-arrow decomposition, but so does every generated strong embedding. That observation alone does not identify them. No counterexample to fullness under the entire *working package is claimed. The designated small subobjects give a colimit in* $\mathbf{K}^{\mathrm{gen}}$ *and in the ambient diagram category; their colimit property for all cocones in* $\mathcal{C}$ *is not established. Mixed-arrow factorization uses all the* H_μ*, not just* H_{μ_0}*.*

Remark 4.20 (Type compatibility, formerly condition $(*)$)**.** *For* $X, Y \in \mathcal{C}_\mu$*,* $Z \in \mathcal{C}_\nu$*,* $\mu \leq \nu$*, any* $f \in H(X, Z)$*,* $g \in H(Y, Z)$*, and bounded* $\mathbf{a}, \mathbf{b}$*,*

$$\exists s : X \cong Y \ (s\mathbf{a} = \mathbf{b}) \quad \Longleftrightarrow \quad \exists u \in \mathrm{Aut}(Z) \ (uf\mathbf{a} = g\mathbf{b}).$$

Indeed, arrow preservation identifies the two outer lifted types with their respective images, and the orbit axiom applies at each level. Thus there is no need to add $(*)$ *separately. The lifting relation is primary; its equivalence and orbit characterization remain substantive axioms. In particular, an* H*-arrow agrees on each bounded tuple with a system isomorphism at the same level. This is not automatic for arbitrary structural or strong embeddings.*

Remark 4.21 (Exhaustion is not an assertion about arbitrary partial maps)**.** *Compatible diagram matches give partial maps* $p_n : A_n \to Y$ *with* $p_n \subseteq p_{n+1}$*. Equality and atomic information make their union a structural embedding once the* A_n *exhaust* X*; fixed constants are preserved when they occur. The partial domains need not be models. The enumerated* B_n^v *must occur* within *the matched lists. An unrelated resolution of each vertex is insufficient to invoke the source criterion: a bounded set can meet all stages of a countable exhaustion without being contained in any one stage.*

The images $p[B_n^v]$ *are nested strong submodels of* $Y(v)$*, and their inclusions are strong by coherence. The chain and common-upper-bound axioms of* $\mathbf{K}$ *show* $p[X(v)] \leq_{\mathbf{K}} Y(v)$*. This proves component strongness, not a possible extra condition on* H*. At the intended countable-cofinality levels, Lemma A.11 also resolves the earlier ordinal-indexing concern for nested lists.*

In fact, a single bounded match from $X \in \mathcal{C}_\mu$ *to* $Y \in \mathcal{C}_\nu$*,* $\mu \leq \nu$*, extends to a whole* H*-arrow: embed* X *into some* $Z \in \mathcal{C}_\nu$ *by a presentation arrow, use type invariance, and then use the orbit axiom to map* Z *isomorphically to* Y *with the prescribed values. Thus the strong-image condition for an enumerated* B_n^v *follows from the full other axioms in this situation. We retain it explicitly to match the source's criterion and its verification obligations. The whole arrows extending different partial matches need not agree outside their prescribed lists; their union is not what is being taken.*

Remark 4.22 (Warnings retained from the reading discussion)**.**

(i) *Formal evaluation of* Ψ *at an arbitrary linear order does not establish membership in* $\mathbf{K}^{\mathrm{gen}}$ *or eligibility. Source Lemma 5.4 concerns padded orders and uses additional arguments. No skeleton or term-label expansion is silently named in the structured-isomorphism relation.*

(ii) *System categoricity is stronger than vertex categoricity. For the walking-arrow shape and pure sets, a bijective internal map and a proper injection are nonisomorphic systems even with all vertices of size* μ*. The source verifies suitable system packages in Proposition 5.6 and Theorems 8.6 and 9.3; Proposition 11.7 assumes the needed geometric-system categoricity, and Proposition 5.8 takes categorical systems as input.*

(iii) *Rigid indexing well-orders do not prevent automorphisms of the system reduct. Matching support patterns are compared through order arrows into a common presentation, not automorphisms of the index order. At one level the orbit axiom then gives a system automorphism.*

(iv) *These lifted types are not identified here with Galois types in the entire generated AEC. No general amalgamation assumption for that AEC is implicit. The comparison with placed diagrams after selected-level amalgamation remains a separate reading question.*

5 Questions for the next reading session

1. Can Theorem 3.11 be isolated as an independent structural theorem for categorical AECs?
2. Once Theorem 4.11 supplies amalgamation at selected levels, how exactly do comparison diagrams relate to conventional Galois types? In particular, distinguish types of tuples with no parameters from types over a model.
3. How does the later finite-system and boundary-lifting calculus compare with Shelah–Vasey multidimensional independence and excellence?
4. Which of the subsequent independence, isolation, and geometry arguments can be translated into standard AEC language without losing content?

6 Simplification ledger and Lean-audit milestones

Result	**Working assessment**
Lemmas 2.1–2.2	Standard; omit from conceptual account.
Lemma 2.5	Erdős–Rado; compress.
Theorem 2.6	Standard EM construction and finite-support functoriality.
Theorem 3.5	Essential order-orbit analysis.
Lemma 3.6	Standard EM functoriality; beware converse.
Lemmas 3.7–3.10	Mostly alignment and ordinal bookkeeping; Proposition 3.9 is substantive.
Theorem 3.11	**First major new theorem: reversal on bounded tuples.**
Lemma 4.1	LS/coherence argument once reversal is available.
Proposition 4.2	**Second major new theorem: stabilization.**
Proposition 4.3	Back-and-forth from stabilized diagrams.
Theorem 4.11	**Third major milestone: amalgamation at selected levels.**

The first stage can be compressed into three conceptual assertions:

$$\boxed{\text{Uniform orbit bound} \implies \text{Bounded-tuple reversal} \implies \text{Stabilized diagrams.}}$$

These also suggest the first nontrivial Lean-verification milestones. Our separate preliminary Lean inspection found foundational AEC definitions and elementary model-existence lemmas in `lean/OAI/ModelTheory/AbstractElementary/ModelExistence.lean`, but did not establish formal verification of the orbit bound, reversal, or stabilization.

A The source working package and the equivalence proof

A.1 Source formulation and comparison conventions

The following is a mathematical restatement, in our notation, of [2, Convention 5.1 and Definition 5.2, pp. 29–30]; it is not a verbatim transcription. The original PDF was checked directly. Its four numbered clauses are preserved here as (S1)–(S4).

Definition A.1 (Source working package)**.** *Fix an unbounded definable eligible class $\mathcal{S}$, a finite system shape, and a set-sized finite-support recipe F. For each $\mu \in \mathcal{S}$, specify a class $\mathcal{C}_\mu$ of systems with each vertex a* $\mathbf{K}$*-model of size μ. Specify arrows and comparisons D on bounded sorted tuples. Systems have their full code of internal maps and fixed named data. Arrows are commuting structural embeddings with strong vertex maps, possibly subject to a further condition.*

(S1) ***Structural arrows and unions.*** *All structured isomorphisms are arrows; arrows compose and are invariant under isomorphism. If $X \subseteq Y \subseteq Z$ are eligible and both inclusions into Z are arrows, so is $X \hookrightarrow Y$. A nonempty increasing chain of inclusion arrows in $\mathcal{C}_\mu$ whose union has total size μ has an eligible union and inclusion arrows. If its stages include by arrows into one eligible object of level at least μ, so does its union.*

(S2) ***Presentations and mixed downward closure.*** *For every well-order L of eligible size μ, $F(L) \in \mathcal{C}_\mu$. Every eligible object is isomorphic to such a presentation at its level. Elements are labels on finite increasing supports; equality, functions, relations, and internal maps are evaluated uniformly from labels and finite order/equality patterns. Substitution along order embeddings gives arrows, including between different eligible levels. If $\nu \leq \mu$ are eligible, every set of at most ν vertex occurrences in a μ-object is contained in a ν-object included by an arrow, retaining the fixed distinguished data.*

(S3) ***Diagram comparisons.*** *For each sorted index pattern, D is an equivalence relation on tuples at the eligible levels where that pattern has length strictly below the level. It preserves the atomic information of the whole system code, is compatible with restriction and reindexing, and is invariant under all arrows. At any one level, equality is exactly equivalence under an isomorphism of the containing systems. There is a set bound on the number of diagrams of each arity, independent of level. A joint diagram extending the diagram of an actual tuple is realizable over that tuple when the joint length is bounded at the receiving level; witnesses may come from larger eligible levels. On term tuples, the comparison depends only on the labels and joint order/equality pattern of supports, also across levels. The empty tuple has one diagram.*

(S4) ***Exhaustion.*** *Compatible matches of increasing bounded sorted lists, exhausting a source object, assemble to an arrow when the lists include nested enumerations of small strong submodels of each source vertex, exhausting that vertex, and each image enumeration is a strong target submodel. The source writes these vertex submodels as V_n. AEC unions and the common-upper-bound axiom establish component strongness; any extra scheme-specific arrow requirement must be checked for this construction when proving the package for that scheme.*

The source's main convention uses strong-limit levels of countable cofinality, apart from explicitly preliminary levels. It does not explicitly quantify an ordinal indexing the opening lists of (S4). Our comparison below handles the written countable interpretation, and also arbitrary ordinal-indexed nested *lists at countable-cofinality source levels. It does not replace chains by arbitrary directed families.*

Remark A.2 (Scope of the equivalence claim)**.** *For the comparison, use the* same *concrete system code on both sides for a nonempty finite system shape, with* $|\tau^{\mathrm{gen}}| \leq \mu_0$ *and* $\mu_0 \geq \mathrm{LS}(\mathbf{K})$*. These bounds normalize the usual AEC convention that its LS number bounds its vocabulary. They are explicit hypotheses of our comparison, not inequalities stated in source Definition 5.2. One can restrict to an eligible tail above these fixed bounds, but doing so changes the minimum and discards preliminary levels; it is not an equivalence preserving the original class of levels.*

For $\mathrm{Diag}_{\mathcal{I}}(\mathbf{K})$*, assume the source's specified internal maps are structural embeddings. Alternatively use the general system category of Definition 4.13, where this restriction is unnecessary. A graph is represented by its free category; relations between paths are retained if they are part of the chosen shape. Neither a new commutativity condition nor additional named skeleton data may be imposed during translation.*

A.2 Consequences of lifted comparison

Lemma A.3 (Common targets and basic type properties)**.** *Under (A1)–(A4), every two eligible objects have an eligible common target. Lifted types have atomic information, restriction and reindexing, and invariance under* H*. The empty type is unique, and every eligible level is categorical as a class of systems.*

Proof. Present X, Y by well-orders L, M. Embed both into a well-order obtained by concatenating them and appending a well-order of sufficiently large eligible size. The induced arrows, transported through the charts, give a common target. For the empty tuple these arrows witness E with the identity automorphism, so its type is unique. The orbit axiom then gives same-level system categoricity.

In a witness $hf\mathbf{a} = g\mathbf{b}$, the maps hf and g are structural embeddings of the full codes. They preserve and reflect atomic formulas, including equalities, sorts, internal maps, and named constants. Thus the tuples have the same atomic information. Restricting or reindexing the tuple keeps the same witness, as long as the resulting arity is bounded. For $f \in H(X, Y)$, the maps f, id_Y and id_Y witness $E((\mathbf{a}; X), (f\mathbf{a}; Y))$. Equivalence then gives invariance of all comparisons under applying arrows. Notice that the direct witness is automatic, whereas transitivity and the orbit characterization are not. □

Lemma A.4 (Mixed downward closure and transport)**.** *Every set* A *of at most* ν *occurrences in* $X \in \mathcal{C}_\mu$*, where* $\nu \leq \mu$ *are eligible, is contained in the image of an arrow from a* $\mathcal{C}_\nu$*-object. Every eligible* X *has an arrow into some object at any prescribed larger eligible level.*

Proof. In a well-ordered chart for X, choose finite supports for the elements of A and, if needed, for the fixed named data. There are at most ν positions, since the code has at most $\mu_0 \leq \nu$ names. Extend them to a suborder J of size ν. Its presentation maps by an arrow into the chart; transporting and taking its image gives the required inclusion. A well-order of cardinality ν has order type in $[\nu, \nu^+)$, as required by the definition of $\mathcal{C}_\nu$. For upward transport append an order of the desired larger cardinality to the chart for X and use its inclusion arrow. □

Lemma A.5 (Uniform bound, realization, and term naturality)**.** *Lifted types satisfy the bound, realization, and naturality clauses of (S3).*

Proof. Fix a sorted arity of cardinality θ, choose eligible $\kappa > \theta$, and fix $M \in \mathcal{C}_\kappa$. Any tuple of that arity can be transported to level κ: use the preceding lemma to shrink its containing object if larger,

or to embed it upward if smaller. Arrow invariance preserves its type, and categoricity takes it into M. Thus there are at most κ^θ types of that sorted arity. This is a bound on equivalence classes, not a claim that a class of all realizations is literally a set.

For realization let $\mathbf{a} \in X \in \mathcal{C}_\mu$ and $(\mathbf{b}, \mathbf{c}) \in Y$ have total length bounded at both levels, with $\mathrm{tp}_{\mathrm{la}}(\mathbf{a}; X) = \mathrm{tp}_{\mathrm{la}}(\mathbf{b}; Y)$. Transport the entire joint tuple into $Y' \in \mathcal{C}_\mu$ by the preceding lemma. Its image $(\mathbf{b}', \mathbf{c}')$ retains its type. The orbit axiom supplies $s : Y' \cong X$ with $s\mathbf{b}' = \mathbf{a}$. Then $\mathbf{d} = s\mathbf{c}'$ realizes the requested joint type over $\mathbf{a}$. The phrase "possible at a larger level" in (S3) is included in this argument; a comparison is only defined where its arity is bounded.

For naturality, identify the joint supports of the two term records by the specified order isomorphism. Amalgamate their two indexing well-orders over those supports: in each cut concatenate the two remaining well-ordered intervals, retaining each common support point. This is again a well-order. Append positions if necessary to give it eligible size. The two induced arrows carry equal labels with their matched supports to literally the same tuple. They witness E with the identity automorphism. This works across different levels and includes supports for named data. It uses the *joint* pattern, not merely separate order types of individual supports. □

A.3 The generated AEC

Lemma A.6 (Small designated decompositions)**.** *Under (S1)–(S2), or under the corresponding alternate axioms, the poset*

$$P_{\mu_0}(X) = \{A \in \mathcal{C}_{\mu_0} : A \hookrightarrow X \text{ is an arrow}\}$$

is μ_0^+-directed and has union X. Its transition inclusions are designated arrows.

Proof. Each occurrence belongs to some member by mixed downward closure. For any family of at most μ_0 members, their total sorted universe has size at most μ_0. Mixed downward closure gives a small designated subobject containing this union. Coherence makes all the inclusions from the original members arrows. This proves μ_0^+-directedness, not merely pairwise directedness, and also the assertion about the union. □

Theorem A.7 (Standard upward completion, applied to the small slice)**.** *Under (S1)–(S2) and the coding bounds of Remark A.2, the AEC in μ_0 $(\mathcal{C}_{\mu_0}, H_{\mu_0})$ has an upward completion $\mathbf{K}^{\mathrm{gen}}$ with LS number μ_0, and*

$$\mathbf{K}^{\mathrm{gen}}_{\mu_0} = (\mathcal{C}_{\mu_0}, H_{\mu_0})$$

on both objects and embeddings. Every eligible object belongs to this AEC, and every designated arrow is a strong embedding in it. Its concrete forgetful functor to the ambient system category is faithful and preserves directed colimits.

Proof. Let $A \leq_0 B$ mean that $A \subseteq B$ and this inclusion belongs to H_{μ_0}. Isomorphism invariance, reflexivity, transitivity, coherence, and unions and smoothness at size μ_0 follow from (S1). Thus this is an AEC in μ_0. Apply the standard upward-completion theorem [1, Definition 1.0.16 and Fact 1.0.18], citing Shelah's Chapter II, Lemma 1.23. It forms directed unions of the small models with coherent small-arrow witnesses for the strong relation; it does not inherit all ambient morphisms. The theorem gives the AEC axioms, LS number, and exact recovery of the small slice. We invoke this established completion theorem rather than claim to reprove it here.

The preceding lemma expresses each eligible X as such a directed union, so $X \in \mathbf{K}^{\mathrm{gen}}$. For an arrow $f : X \to Y$, identify X with $f[X] \subseteq Y$. For every $A \in P_{\mu_0}(X)$ its inclusion into Y is an

arrow by composition. Whenever such A is contained in $B \in P_{\mu_0}(Y)$, coherence gives $A \leq_0 B$. Consequently the two small-subobject presentations fit together as the coherent presentations defining $X \leq_{\mathrm{gen}} Y$: one may use $P_{\mu_0}(X)$ inside $P_{\mu_0}(Y)$, with the inherited small-arrow order. This also gives, equivalently, directed pairs of small source and target pieces covering the whole arrow.

Every small map is componentwise strong. Directed unions and the common-upper-bound property in $\mathbf{K}$ therefore make all the induced vertex maps strong in the completion; internal symbols and fixed names are retained on the sorted unions. Colimits in this AEC and in the ambient system category are computed on the same carriers. Thus the forgetful functor is well-defined, concrete, faithful, and preserves directed colimits. In the diagram version, unions also retain the internal structural-embedding property. □

Remark A.8 (What an arbitrary ambient completion does not give)**.** *The completion theorem has no conclusion that all generated embeddings between eligible objects satisfy their original geometric conditions. The inclusion $H(X,Y) \subseteq \mathrm{Emb}_{\mathbf{K}^{\mathrm{gen}}}(X,Y)$ is all the translation needs. Conversely, any AEC as in (D1) with the prescribed small slice is recovered from that slice by upward completion: its small strong submodels cover every model by LS and coherence, and coherent small-map unions are strong by the directed-union theorem for AECs. Thus the completion is an auxiliary presentation, not a freely chosen source of additional eligible arrows.*

A.4 The two translations

Lemma A.9 (The source relation is forced)**.** *For a source working package, D-equality is exactly lifted automorphism comparison E formed from its eligible objects and designated arrows. In particular, there is no additional choice of cross-level equivalence once those objects and arrows are fixed and the package axioms hold.*

Proof. If $hf\mathbf{a} = g\mathbf{b}$ witnesses E, invariance under the arrows f, g, h and transitivity of D give $D(\mathbf{a}; X) = D(\mathbf{b}; Y)$. Conversely, choose any common eligible target Z for X, Y, with arrows f, g, using their well-ordered presentations. Arrow invariance gives $D(f\mathbf{a}; Z) = D(g\mathbf{b}; Z)$. The same-level orbit clause supplies $h \in \mathrm{Aut}(Z)$ with $hf\mathbf{a} = g\mathbf{b}$, witnessing E. The arity is bounded in Z because it was bounded in the source objects. Since D is an equivalence relation, so is E. □

Lemma A.10 (Short and arbitrary size-preserving fixed-level chains)**.** *The chain clauses in (A3) imply those in (S1).*

Proof. Let $(X_i)_{i<\delta}$ be a nonempty inclusion chain at level μ whose union has size μ. If it has a final stage, that is the union. If $\mathrm{cf}(\delta) \leq \mu$, use a cofinal subchain of length at most μ, apply (A3), and use composition for omitted stages. If $\mathrm{cf}(\delta) > \mu$, choose a stage containing each occurrence of the union. Those at most μ stage indices have a common bound below δ, so the chain has already stabilized to its union. The common-upper-bound conclusion follows by the same argument. The converse follows immediately by taking chains of length $< \mu^+$. □

Lemma A.11 (Countable reduction of nested exhaustion)**.** *Suppose $\mathrm{cf}(\mu) = \omega$, $|X| = \mu$, and $(A_\alpha)_{\alpha<\delta}$ is an increasing family of sets of size $< \mu$ with union X. Then δ has countable cofinality, and a cofinal countable subchain still exhausts X. Any simultaneously nested vertex resolutions and compatible matches restrict to this subchain.*

Proof. There can be no final stage. Put $\kappa = \mathrm{cf}(\delta)$ and pass to a cofinal subchain. If $\kappa > \mu$, the indices of the first occurrences of all elements have a bound below δ, contradicting boundedness of a stage. Thus $\kappa \leq \mu$. If $\mu = \aleph_0$, it follows that $\kappa = \omega$. Otherwise μ is singular, so the regular cardinal κ is $< \mu$. The union has size μ, hence the cardinalities of the stages are unbounded below μ; a bound $\lambda < \mu$ would give union size at most $\kappa \cdot \lambda < \mu$ (enlarge λ to be infinite). Choose countably many stages whose sizes are cofinal in μ. If $\kappa > \omega$, their indices have a bound below δ, forcing one stage to have size at least μ, a contradiction. Hence $\kappa = \omega$. Cofinal restriction retains all the required containments, matches, strong images, and unions. □

Theorem A.12 (Equivalence on the specified scope)**.** *Fix the concrete coding and cardinal bounds of Remark A.2. At eligible levels of countable cofinality, the source package of Definition A.1 and the alternate definition of Definition 4.15 are equivalent, preserving the eligible objects, designated arrows, presentation recipe, and all tuple comparisons. Use the diagram version when internal maps are structural embeddings, or the general system version otherwise. Interpret (S4) for countable or, equivalently here, ordinal-indexed nested exhaustion with the corresponding nested vertex witnesses.*

Proof. Source to alternate. Theorem A.7 constructs $\mathbf{K}^{\mathrm{gen}}$ and embeds all the eligible data in it, recovering the small slice exactly. Use F as the finite-support system-code recipe Ψ; no all-order AEC blueprint theorem is required. By (S2) the eligible objects are exactly the isomorphism closures in (D3). Retain the original H. Lemma A.9 identifies D with E, proving (A1) and (A4). Presentation arrows give (A2). Structural closure gives (A3), and (S4) gives (A5) upon taking the shared enumerations and their partial maps $p \restriction A_n$.

Alternate to source. Forget the auxiliary completion, retain $\mathcal{C}, H, F$, and define D-equality to be equality of lifted types. The structural assertions of (S1) follow from (A3) and Lemma A.10. The presentation assertions of (S2) are (D2)–(D3) and (A2); its mixed downward part is Lemma A.4. Lemmas A.3 and A.5, together with (A1) and (A4), prove every clause of (S3), including the empty type and the uniform bound.

For (S4), compatible matches define well-defined sorted partial injections because they preserve equality. Their union is total on X. Every finite list of arguments of a function or relation eventually lies in one stage; atomic agreement there shows preservation and reflection in the union. Function values also eventually occur, and names are preserved. Thus the union is a structural embedding of the whole system code. Its vertex witnesses and their strong images are exactly those in (A5), so it is in H. If the exhaustion is indexed by an arbitrary ordinal, apply Lemma A.11 first. Conversely countable exhaustion is a particular such exhaustion.

The translations leave objects and H unchanged, and Lemma A.9 shows that they recover the original D. The completion is canonical in the sense explained after Theorem A.7. This proves equivalence of the packages, not higher-level fullness of H in the completion. □

A.5 What remains open or outside this equivalence

1. The exact diagram formulation does not cover non-embedding internal maps without the stated ambient-category change. This is a scope difference, not an unproved implication that we silently assume.

2. The language and LS bounds at the original minimum are explicit normalizations. If they fail for a source descriptor, the theorem applies to a suitable eligible tail, not to all its preliminary levels unchanged.

3. Countable reduction resolves the indexing issue for nested ordinal chains at countable-cofinality levels. It says nothing about arbitrary directed matches or preliminary source levels of other cofinalities; a stronger exhaustion convention there must be added on both sides to obtain the same comparison.

4. Whether the full package forces all eligible generated strong embeddings to lie in H remains open in these notes. It is neither assumed nor needed in the equivalence proof.

5. Formal arbitrary-order evaluations remain distinct from actual AEC membership. Also, changing the eligible class by later closure refinements requires the source's separate supply-of-levels hypotheses; mere unboundedness is not closedness under those refinements.

References

[1] Adi Jarden and Saharon Shelah. Non-forking frames in abstract elementary classes. https://shelah.logic.at/files/182850/875.pdf, 2013. Sh:875, version 2013-03-03; Definition 1.0.16 and Fact 1.0.18, citing Shelah, Chapter II, Lemma 1.23.

[2] OpenAI. Eventual categoricity conjecture manuscript. https://github.com/openai/math/blob/main/preprints/Eventual-Categoricity-for-Abstract-Elementary-Classes-September-24-2026/paper.pdf. September 24, 2026 manuscript; working-package definition on pp. 29–30.